30 days of simplydata

An hour a day: a warm-up from memory, a lesson, and a task to do yourself

Version 0.1.0 • two tracks of 30 days • made from the code by `docs/make_30_days.py`

How to use this guide

The same 30 days are in Python. Start once, then type one word each day:

```
import simplydata as sdata
sdata.start_30_days()                    # the Python track: you already write some Python
sdata.start_30_days(track="beginner")    # the beginner track: you have never programmed
sdata.today()                            # every day: the day to do next
sdata.answers()                          # the answers to the warm-up and the task
```

This guide is the paper version, to keep next to the manual (`simplydata_manual.pdf`): every page number is a link that opens the manual at that page. Each day has three parts and takes about an hour:

1. **Warm-up, 5 minutes.** A few questions about words from earlier days. Answer them *from memory*, on the lines, before you look anything up. Then check the answers at the back of the guide (each day has a link to them) or with `sdata.answers(n)`.
2. **What to do, about 45 minutes.** The lesson to work through (`sdata.lesson(n)`), the pages of the manual to read and the words to look up with `sdata.help("word")`. Type every example yourself: typing it, and getting it wrong, teaches more than copying.
3. **Try it yourself, 10 minutes.** A task to do without looking back. If yours runs and answers the question, it is right; the worked answers are at the back.

Tick the boxes at the top of the page as you go, and the day on the calendar of your track. **Missed a day, or a week?** That is fine. Carry on where you stopped: the days build on each other, and nothing is lost by taking a break. In Python, `sdata.today()` does the same.

Why a warm-up from memory?

Trying to remember something fixes it in memory better than reading it again: in experiments, people who were tested on a text remembered more of it days later than people who read it again (Roediger and Karpicke 2006). Practice spread over several days also works better than the same practice in one go (Cepeda et al. 2006). So each warm-up asks one question about yesterday, one about three days ago and one about a week ago (and, later on, two weeks ago). Review days (7, 14, 21 and 28) ask about the whole course, and the questions change each time a day comes back.

Which track?

Python track	For people who already write some Python. About two lessons a day for 20 days, with review days on days 7, 14, 21 and 28, then an eight-day project of your own (days 21 to 28), a final test from memory and a look at the same work in pandas and matplotlib. Start: `sdata.start_30_days()`.
Beginner track	For people who have never programmed. Three days of just enough Python first, then about one lesson a day, with review days on days 7, 14 and 21, a small project of your own (days 27 and 28), a final test from memory and a look back. Start: `sdata.start_30_days(track="beginner")`.

The words in Python

sdata `sdata.start_30_days()` p. 239	Start 30 days of simplydata: an hour a day of warm-up questions, lessons and tasks.
sdata `sdata.today()` p. 240	Show today's day of the 30 days of simplydata: warm-up questions, what to do and a task.
sdata `sdata.day()` p. 240	Show any day of the 30 days, without changing where you are.
sdata `sdata.answers()` p. 240	The answers to a day's warm-up questions, and a worked answer to its task.
sdata `sdata.next_day()` p. 241	Go on to the next day now, without waiting for tomorrow.

Your progress is kept only on your computer, in a small file (`~/.simplydata/progress.json`). Nothing is sent anywhere.

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References

Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: taking memory tests improves long-term retention. *Psychological Science*, 17(3), 249–255. https://doi.org/10.1111/j.1467-9280.2006.01693.x

Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: a review and quantitative synthesis. *Psychological Bulletin*, 132(3), 354–380. https://doi.org/10.1037/0033-2909.132.3.354

Python track

For people who already write some Python

About two lessons a day for 20 days, with review days on days 7, 14, 21 and 28, then an eight-day project of your own (days 21 to 28), a final test from memory and a look at the same work in pandas and matplotlib. Tick each day when you have done it; the day's number is a link to its page.

	Day	What you do	Lesson	Page
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1

Day 1 of 30 • Python track

Your first chart and your first table

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Make a bar chart with titles and labels, and a first table you keep rows of and sum up. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

Nothing to remember yet on day 1: just start.

2 What to do

- Work through `sdata.lesson(1)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(1, solutions=True)`.
- Work through `sdata.lesson(2)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(2, solutions=True)`.
- Run the example below: your first table. A table and a chart are separate things: one clean table can feed many charts. Tables get their own lessons on day 6.

In the manual lesson 1, p. 10; lesson 2, p. 13; "Welcome", p. 2.

Words to look up sdata `sdata.example()` p. 111; sdata `sdata.chart()` p. 111; chart `chart.bars()` p. 113; chart `chart.save()` p. 112; chart `chart.title_top()` p. 131; chart `chart.label_x()` p. 132; chart `chart.label_y()` p. 132; chart `chart.rotate_labels_x()` p. 138.

Example: type it yourself

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("marks", above=50)
table.average_of("marks", for_each="class")
chart = table.chart()
chart.bars(x="class", y="marks")
chart.save("good_marks.png")
```

What each line does

1. Load simplydata.
2. Start a table from the marks practice table.
3. Keep only the rows with marks above 50. It says how many rows are left.
4. The average mark of each class, in a sentence.
5. Start a chart from the table as it is now.
6. One bar for each class.
7. Save it.

3 Try it yourself, without looking back

1. Make a bar chart of the `"weather"` table with `x="city"` and `y="rain"`, and save it as `"rain.png"`.
2. Draw dots of `"hours"` (x) against `"marks"` (y) from the marks table, with a title, a name for each axis and a caption at the bottom.
3. Start a table of the weather, keep the rows with `"temperature"` above 15, and give the average `"rain"` for each city.

Worked answers: page 67, or `sdata.answers(1)`.

2

Day 2 of 30 • Python track

Colours, word families and saving

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Give groups their own colours, learn how simplydata words are built, and save in every file type. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word lists the practice tables? (day 1)
2. How do you start a chart of a table called marks? (day 1)
3. Which word draws one bar for each name, as tall as the marks? (day 1)

Answers: page 67, or `sdata.answers(2)`.

2 What to do

- Work through `sdata.lesson(3)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(3, solutions=True)`.
- Read `"How the words are built"` in the manual (two pages). It is the key to guessing words you have never seen.
- Type `sdata.words()` and look for the families: `keep_` and `remove_`, …_x and …_y, `add_`…_column, …_of.
- Then do the task: guess first, check with `sdata.help` after.

In the manual lesson 3, p. 16; "How the words are built", p. 5.

Words to look up chart `chart.colour()` p. 140; chart `chart.colours()` p. 141; chart `chart.legend_bottom()` p. 144; chart `chart.order_x()` p. 146; chart `chart.save()` p. 112; chart `chart.hide_legend()` p. 145; chart `chart.legend_left()` p. 144; sdata `sdata.words()` p. 238.

3 Try it yourself, without looking back

1. Weather table: draw lines of `"temperature"` over `"month"`, one colour for each city. Put the legend on top and use the style `"paper"`.
2. Save one chart as `.png`, `.pdf` and `.svg`. Then guess, before you look anything up: which word hides the legend, and which puts it on the left? Try your guesses on a chart with `colour=`.

Worked answers: page 67, or `sdata.answers(2)`.

3

Day 3 of 30 • Python track

Every kind of chart

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Draw lines, dots, histograms, boxes, violins, pies and areas, and know when to use each. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you give each class its own colour in a bar chart? (day 2)
2. Which word moves the legend below the chart? (day 2)
3. How do you save a chart as a PDF? (day 1)

Answers: page 68, or `sdata.answers(3)`.

2 What to do

- Work through `sdata.lesson(4)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(4, solutions=True)`.

In the manual lesson 4, p. 21.

Words to look up chart `chart.lines()` p. 113; chart `chart.dots()` p. 114; chart `chart.histogram()` p. 115; chart `chart.boxes()` p. 116.

3 Try it yourself, without looking back

1. Draw violins of `"temperature"` for each city in the weather table, with all the dots on top.

Worked answers: page 68, or `sdata.answers(3)`.

4 Statistics on charts, and axes

Day 4 of 30 • Python track

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Show averages with error bars and fit lines, and control the axes with ranges and reference lines. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which chart shows how the values of one column are spread, in bins? (day 3)
2. Which word puts a title above the chart? (day 1)
3. Which two words show the spread of each group, with every value as a dot on top? (day 3)

Answers: page 68, or `sdata.answers(4)`.

2 What to do

- Work through `sdata.lesson(5)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(5, solutions=True)`.
- Work through `sdata.lesson(6)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(6, solutions=True)`.

In the manual lesson 5, p. 26; lesson 6, p. 31; "Standard deviation or standard error?", p. 244.

Words to look up option `show=` p. 121; option `error=` p. 121; chart `chart.numbers_on_bars()` p. 133; chart `chart.fit_line()` p. 122; chart `chart.range_y()` p. 136; chart `chart.line_at_y()` p. 130; chart `chart.shade_between_y()` p. 131; chart `chart.log_scale_y()` p. 136.

3 Try it yourself, without looking back

1. Weather table: draw bars of the average `"rain"` for each city with 95 % confidence intervals, and write the numbers on the bars.
2. Growth table: draw lines of `"population"` over `"year"`, one for each country, with a line at x = 2010 and the y-axis starting at 0.

Worked answers: page 68, or `sdata.answers(4)`.

5

Day 5 of 30 • Python track

Several charts, and a figure for a journal

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Put charts side by side or in a grid, and save a figure at the size and resolution a journal asks for. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which option makes a bar show the median instead of the average? (day 4)
2. Which word lets you choose the colours of the groups? (day 2)
3. Which option adds error bars that say how sure we are of the average? (day 4)

Answers: page 69, or `sdata.answers(5)`.

2 What to do

- Work through `sdata.lesson(7)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(7, solutions=True)`.
- Work through `sdata.lesson(8)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(8, solutions=True)`.

In the manual lesson 7, p. 36; lesson 8, p. 41.

Words to look up chart `chart.one_chart_per()` p. 148; sdata `sdata.side_by_side()` p. 148; sdata `sdata.one_above_other()` p. 149; sdata `sdata.grid()` p. 149; chart `chart.second_y()` p. 139; chart `chart.size()` p. 142; chart `chart.style()` p. 142; chart `chart.note()` p. 133.

3 Try it yourself, without looking back

1. Make a histogram and a box chart of the marks, and put them one above the other on one page.
2. A figure for a journal: dots of `"hours"` against `"marks"`, style `"paper"`, 8.5 cm wide and 7 cm high, font size 8, saved as a `.tif` with 600 dpi.

Worked answers: page 69, or `sdata.answers(5)`.

6

Day 6 of 30 • Python track

Open a table, choose rows and columns

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Look at a table, keep and remove rows and columns (by value and by position), sort, and undo. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word draws one small chart for each class? (day 5)
2. Which word draws one dot for each row, x against y? (day 3)
3. Which word puts two charts next to each other on one page? (day 5)

Answers: page 69, or `sdata.answers(6)`.

2 What to do

- Work through `sdata.lesson(9)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(9, solutions=True)`.
- Work through `sdata.lesson(10)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(10, solutions=True)`.

In the manual lesson 9, p. 48; lesson 10, p. 52.

Words to look up sdata `sdata.table()` p. 151; table `table.show_first_rows()` p. 152; table `table.describe_columns()` p. 153; table `table.count_rows()` p. 176; table `table.keep_rows()` p. 157; table `table.keep_columns()` p. 155; table `table.sort_by()` p. 169; table `table.undo()` p. 211.

3 Try it yourself, without looking back

1. Open the `"shop"` practice table. How many rows does it have, and how many rows does each shop have?
2. Weather table: keep the city `"Sunnyvale"`, keep only the columns month, temperature and rain, sort by rain with the biggest first, and save it as `"sunny.csv"`.

Worked answers: page 69, or `sdata.answers(6)`.

7

Day 7 of 30 • Python track

Review of the first week

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Remember the words of days 1 to 6 without looking, and use them together in one program. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you look at the first 5 rows of a table? (day 6)
2. How do you name the y-axis? (day 1)
3. Which word draws a straight line through the dots? (day 4)
4. How do you put the bars in your own order? (day 2)
5. How do you put four charts in two rows, with the letters A to D? (day 5)
6. Which word draws a line with a dot at each point? (day 3)

Answers: page 70, or `sdata.answers(7)`.

2 What to do

- Answer the warm-up on paper, from memory, before you look anything up.
- For each question you could not answer, open its day with `sdata.day(n)` and run its example again.
- Then do the mixed task without looking back. It uses words from the whole week.

3 Try it yourself, without looking back

1. Marks table: keep only class `"A"`, draw bars of the average `"marks"` for each subject with standard error bars, give the chart a title and save it as a PDF.

Worked answers: page 70, or `sdata.answers(7)`.

8

Day 8 of 30 • Python track

Clean the data and add columns

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Find and fill empty cells, fix text and numbers, and add new columns, groups and dates. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word describes each column: its kind, empty cells and numbers? (day 6)
2. Which word adds a second y-axis on the right? (day 5)
3. The names under the bars bump into each other. Which word helps? (day 1)

Answers: page 70, or `sdata.answers(8)`.

2 What to do

- Work through `sdata.lesson(11)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(11, solutions=True)`.

In the manual lesson 11, p. 57; "Missing values and imputation", p. 248.

Words to look up table `table.count_empty_cells()` p. 163; table `table.fill_empty_cells()` p. 164; table `table.make_numbers()` p. 165; table `table.add_column()` p. 170.

3 Try it yourself, without looking back

1. Shop table: count the empty cells, fill the empty `"sales"` with the line between neighbours for each shop (sorted by `"date"`), and add a column `"weekday"` with the name of the weekday.

Worked answers: page 70, or `sdata.answers(8)`.

9

Day 9 of 30 • Python track

Summaries, numbers and grids

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Sum up columns for each group, join and reshape tables, and work with lists and grids of numbers. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word counts the empty cells in each column? (day 8)
2. Which word counts the rows of each fruit? (day 6)
3. If `keep_rows` keeps rows, what would you guess takes columns away? (day 2)

Answers: page 71, or `sdata.answers(9)`.

2 What to do

- Work through `sdata.lesson(12)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(12, solutions=True)`.
- Work through `sdata.lesson(13)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(13, solutions=True)`.

In the manual lesson 12, p. 64; lesson 13, p. 72.

Words to look up table `table.average_of()` p. 177; table `table.summary_of()` p. 179; table `table.correlation_between()` p. 182; table `table.join_with()` p. 209; sdata `sdata.numbers()` p. 214; sdata `sdata.random_numbers()` p. 214; nums `nums.median()` p. 226; nums `nums.percentile()` p. 228.

3 Try it yourself, without looking back

1. Fruit table: the total `"count"` of each fruit, drawn as bars with the numbers on the bars.
2. Make 200 random heights with average 150 and standard deviation 8 (`seed=1`). What is their 90th percentile? Draw them as a histogram.

Worked answers: page 71, or `sdata.answers(9)`.

10

Day 10 of 30 • Python track

Is a difference real? Tests and effect sizes

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Test whether averages differ, find which groups differ, and say how big the difference is. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word gives the average of a column for each group? (day 9)
2. Which word counts the rows? (day 6)
3. Which chart shows how the values of one column are spread, in bins? (day 3)

Answers: page 71, or `sdata.answers(10)`.

2 What to do

- Work through `sdata.lesson(14)`, examples 1 and 3: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(14, solutions=True)`.
- Read `"The null hypothesis and the p-value"` and `"Effect sizes"` in the manual.

In the manual lesson 14, p. 76; "The null hypothesis and the p-value", p. 245; "Effect sizes: how big is the difference?", p. 246.

Words to look up table `table.compare_averages()` p. 183; table `table.compare_each_pair()` p. 183; table `table.fit_line()` p. 188.

3 Try it yourself, without looking back

1. School table: do the average marks differ between pupils who had breakfast and pupils who did not? Write down the difference, the effect size and the p-value.

Worked answers: page 71, or `sdata.answers(10)`.

11

Day 11 of 30 • Python track

Counts, outliers and many tests

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Test counts of categories, check shapes and outliers, and correct the p-values of many tests. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word tests whether the average marks differ between the classes? (day 10)
2. How do you fill the empty marks with the median? (day 8)
3. Which word checks by eye whether values are bell-shaped? (day 4)

Answers: page 72, or `sdata.answers(11)`.

2 What to do

- Work through `sdata.lesson(14)`, examples 2, 4 and 5: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(14, solutions=True)`.
- Read `"Many tests at once"` in the manual.

In the manual lesson 14, p. 76; "Many tests at once", p. 246.

Words to look up table `table.compare_counts()` p. 185; table `table.show_rows_with_outliers()` p. 162; table `table.compare_shape_to()` p. 185; table `table.add_adjusted_p_value_column()` p. 186.

3 Try it yourself, without looking back

1. School table: is the share of pupils who `"passed"` different with and without breakfast?

Worked answers: page 72, or `sdata.answers(11)`.

12

Day 12 of 30 • Python track

Choosing a test, and writing it up

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Choose the right test with four questions, and write a results sentence a reader can check. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word tests whether the counts of categories differ between groups? (day 11)
2. Which word tells whether two columns go up and down together? (day 9)
3. Which word sets the size of a figure in cm? (day 5)

Answers: page 72, or `sdata.answers(12)`.

2 What to do

- Read `"Which test when?"` and `"Writing it up"` in the manual. Answer its four questions for each task below before you run anything.
- Run `sdata.help("compare_before_after")` and `sdata.help("compare_average_to")` and type their examples.

In the manual "Which test when?", p. 249; "Writing it up", p. 250.

Words to look up table `table.compare_average_to()` p. 184; table `table.compare_before_after()` p. 184; table `table.compare_percent_to()` p. 187.

3 Try it yourself, without looking back

1. Marks table: is each child's Maths mark different from the same child's Science mark? Choose the right word first.
2. School table: is the average `"sleep"` different from 8 hours?

Worked answers: page 72, or `sdata.answers(12)`.

13

Day 13 of 30 • Python track

A whole analysis

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Go from a question and a file to a tested answer and a figure with panels A and B. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. The same children were measured twice. Which word compares the two? (day 12)

2. Which word says which pairs of groups differ? (day 10)

3. Which word keeps only the rows where subject is Maths? (day 6)

Answers: page 72, or `sdata.answers(13)`.

2 What to do

- Work through `sdata.lesson(15)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(15, solutions=True)`.

In the manual lesson 15, p. 81.

Words to look up table `table.steps()` p. 211; sdata `sdata.grid()` p. 149; table `table.save()` p. 212; table `table.as_pandas()` p. 212.

3 Try it yourself, without looking back

1. From memory: weather table, compare the average `"temperature"` of the cities, draw boxes with all the dots, and save the figure as a PDF and the table as `"weather.xlsx"`.

Worked answers: page 72, or `sdata.answers(13)`.

14 Review of the second week

Day 14 of 30 • Python track

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Remember the table and test words of days 6 to 13 without looking, and use them together. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word lists every change you made to a table, for your methods? (day 13)
2. Which word makes all the text on a chart bigger or smaller? (day 1)
3. A word for one axis ends in ...? (day 2)
4. Which word turns a column of text such as `"144"` into numbers? (day 8)
5. Which word sets the y-axis from 0 to 100? (day 4)
6. Which word shows the rows with outliers? (day 11)
7. Which word joins two tables on a column they share? (day 9)
8. How do you save a picture sharp enough for a journal? (day 5)

Answers: page 73, or `sdata.answers(14)`.

2 What to do

- Answer the warm-up on paper, from memory, before you look anything up.
- For each question you could not answer, open its day with `sdata.day(n)` and run its example again.
- Then do the mixed task without looking back.

3 Try it yourself, without looking back

1. School table: count the empty cells, remove the rows with an empty `"sleep"`, give the average sleep for each value of `"passed"`, test the difference, and draw boxes with all the dots.

Worked answers: page 73, or `sdata.answers(14)`.

15 Models: explain one column from several

Day 15 of 30 • Python track

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Explain one column from several others: multiple, logistic, Poisson and mixed regression. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you give the panels of a figure the letters A and B? (day 13)
2. Which word compares one average with a number you choose? (day 12)
3. Which word adds a new column made from other columns? (day 8)
4. How do you start a table from the marks practice table? (day 1)

Answers: page 74, or `sdata.answers(15)`.

2 What to do

- Work through `sdata.lesson(16)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(16, solutions=True)`.

In the manual lesson 16, p. 86.
Words to look up table `table.explain()` p. 200; table `table.add_predicted_column()` p. 200.

3 Try it yourself, without looking back

1. School table: explain `"days_absent"` using `"sleep"` and `"breakfast"` with `outcome="counts"`. Does breakfast matter?

Worked answers: page 74, or `sdata.answers(15)`.

16

Day 16 of 30 • Python track

Patterns across many columns

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Find patterns across many columns with principal components, clusters, NMDS and PERMANOVA. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word explains marks from hours, sleep and breakfast together? (day 15)
2. Which word hands a table on to pandas? (day 13)
3. Which word gives several summaries at once? (day 9)
4. Which file endings give a picture that stays sharp at any size? (day 2)

Answers: page 74, or `sdata.answers(16)`.

2 What to do

- Work through `sdata.lesson(17)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(17, solutions=True)`.

In the manual lesson 17, p. 90.

Words to look up table `table.add_principal_components()` p. 201; table `table.add_clusters()` p. 203; table `table.add_nmds()` p. 204; table `table.compare_profiles()` p. 205.

3 Try it yourself, without looking back

1. Meadow table: add 3 clusters and count the rows of each cluster across `"field"`.

Worked answers: page 74, or `sdata.answers(16)`.

17

Day 17 of 30 • Python track

Survival, and planning a study

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Draw and test survival curves, and work out how many values a study needs. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word adds the principal components PC1 and PC2? (day 16)
2. Which word lists every change you made to a table, for your methods? (day 13)
3. A p-value answers `"could it be chance?"`. What answers `"how big is it?"` (day 10)
4. Which two words show the spread of each group, with every value as a dot on top? (day 3)

Answers: page 74, or `sdata.answers(17)`.

2 What to do

- Work through `sdata.lesson(18)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(18, solutions=True)`.

In the manual lesson 18, p. 95; "Two kinds of mistake, and power", p. 246.

Words to look up table `table.survival_over_time()` p. 205; chart `chart.survival_curves()` p. 207; table `table.compare_survival()` p. 206; sdata `sdata.how_many_needed()` p. 207.

3 Try it yourself, without looking back

1. How many values does each group need to find a large difference (`effect_size=0.8`) with power 0.8? And with a small one (0.2)?

Worked answers: page 74, or `sdata.answers(17)`.

18 Fitting lines, curves and formulas

Day 18 of 30 • Python track

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Fit straight lines, curves and formulas of your own to a table, and draw them on a chart. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word draws survival curves? (day 17)
2. How do you explain a yes-or-no column such as `"passed"`? (day 15)
3. Which word checks whether values are bell-shaped (normal)? (day 11)
4. Which word draws a line across at y = 50, with a label? (day 4)

Answers: page 75, or `sdata.answers(18)`.

2 What to do

- Work through `sdata.lesson(19)`, examples 1 and 2: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(19, solutions=True)`.

In the manual lesson 19, p. 99; "Fitting a line or a formula", p. 247.
Words to look up table `table.fit_line()` p. 188; table `table.fit_curve()` p. 189; table `table.fit_formula()` p. 189; chart `chart.fit_formula()` p. 123.

3 Try it yourself, without looking back

1. Growth table: fit the formula `"forest = a + b * (year - 2000)"` for each country. Which country loses forest fastest?

Worked answers: page 75, or `sdata.answers(18)`.

19

Day 19 of 30 • Python track

Finding, solving and distributions

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Find where a formula is smallest or biggest, solve equations, and work out chances from a distribution. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word fits a straight line on a table? (day 18)
2. Which word puts rows that are alike into groups? (day 16)
3. What should a results sentence give? (day 12)
4. Which word writes a note with an arrow on the chart? (day 5)

Answers: page 75, or `sdata.answers(19)`.

2 What to do

- Work through `sdata.lesson(19)`, example 3: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(19, solutions=True)`.
- Work through `sdata.lesson(20)`, examples 1 and 2: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(20, solutions=True)`.

In the manual lesson 19, p. 99; lesson 20, p. 104; "The shape of the values: distributions", p. 244.

Words to look up sdata `sdata.find_smallest()` p. 190; sdata `sdata.find_biggest()` p. 191; sdata `sdata.solve()` p. 191; sdata `sdata.distribution()` p. 193; dist `dist.chance()` p. 195; dist `dist.percentile()` p. 196; table `table.fit_distribution()` p. 194.

3 Try it yourself, without looking back

1. A box has a square bottom of side x and a height of 10 - x, so its volume is x * x * (10 - x). For which x between 0 and 10 is the volume biggest?
2. Heights are normal with average 150 and standard deviation 8. What share is taller than 165? Which height do 95 % of them stay below?

Worked answers: page 75, or `sdata.answers(19)`.

20

Day 20 of 30 • Python track

Values between points

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Estimate values between the points you measured, and fill an even map for a contour chart. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word finds where a formula is smallest? (day 19)
2. Which word tests whether survival differs between groups? (day 17)
3. How do you give the panels of a figure the letters A and B? (day 13)
4. Which word sorts with the biggest marks first? (day 6)

Answers: page 76, or `sdata.answers(20)`.

2 What to do

- Work through `sdata.lesson(20)`, example 3: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(20, solutions=True)`.

In the manual lesson 20, p. 104.
Words to look up table `table.values_at()` p. 192; table `table.map_of()` p. 193.

3 Try it yourself, without looking back

1. Weather table: estimate the `"rain"` of each city at `month_number` 2.5 and 7.5, joined by straight lines.

Worked answers: page 76, or `sdata.answers(20)`.

21

Day 21 of 30 • Python track

Review, and the question of your project

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Remember the words of the third week, then choose the question and the raw file of your own project. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word estimates a column between the points you measured? (day 20)
2. Which word cuts a column of numbers into named groups? (day 8)
3. Which word keeps only the rows with marks above 50? (day 1)
4. How do you make simplydata numbers from a list? (day 9)
5. Which option gives the rank-based test, for small groups that are not bell-shaped? (day 10)
6. Which word adds the values a model predicts as a new column? (day 15)
7. You did many tests at once. Which word corrects the p-values? (day 11)
8. A few values are much bigger than the rest. Which word helps? (day 4)
9. How do you fit a curve with x and x squared? (day 18)

Answers: page 76, or `sdata.answers(21)`.

2 What to do

- Answer the warm-up on paper, from memory, before you look anything up.
- For each question you could not answer, open its day with `sdata.day(n)` and run its example again.
- Choose data of your own: a file from your work, your studies or a public data set. No data? Use the `"school"` practice table, as the answers do.
- Write your question in one sentence, for example: do pupils who sleep more get better marks?
- Keep the raw file as it is: never change it. All the cleaning will be steps on a table, which you can repeat and undo.
- Open the file with `sdata.table` and look at it.

In the manual "Tidy data: long and wide tables", p. 248.
Words to look up sdata `sdata.table()` p. 151; table `table.describe_columns()` p. 153; table `table.save()` p. 212.

3 Try it yourself, without looking back

1. Trial table: test whether survival differs between the treatments, then explain the risk of dying from `"treatment"` and `"age"` together.
2. Write your question in one sentence. Open your raw file and write down how many rows and columns it has, and which columns have empty cells.

Worked answers: page 76, or `sdata.answers(21)`.

22

Day 22 of 30 • Python track

Project: clean

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Clean your raw file into a tidy table, with every step written down. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word tests whether groups differ in their make-up across many columns (PERMANOVA)? (day 16)
2. Which word finds where the two sides of an equation are equal? (day 19)
3. Which word says how many values you need in each group? (day 17)
4. You made a mistake in the last step. Which word takes it back? (day 6)

Answers: page 77, or `sdata.answers(22)`.

2 What to do

- Project day 2: clean. Go column by column through what `describe_columns` showed you.
- Fix text with spaces or mixed capitals, numbers stored as text, odd codes such as -999, and decide what to do with empty cells (remove, fill, or keep and let the tests skip them).
- Save the clean table under a new name, and keep the `steps()`.

In the manual lesson 11, p. 57.

Words to look up table `table.count_empty_cells()` p. 163; table `table.remove_extra_spaces()` p. 167; table `table.replace_values()` p. 167; table `table.fill_empty_cells()` p. 164.

3 Try it yourself, without looking back

1. Clean your table and save it as a new file. Write one line for each decision you made about empty cells.

Worked answers: page 77, or `sdata.answers(22)`.

23

Day 23 of 30 • Python track

Project: explore

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Explore your clean table with summaries and charts before any test. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word fills an even map from scattered points, ready for a contour map? (day 20)
2. Which word fits a formula of your own, such as growth? (day 18)
3. Which word gives the middle value of numbers? (day 9)
4. Which word shows the numbers on the y-axis as percents? (day 4)

Answers: page 77, or `sdata.answers(23)`.

2 What to do

- Project day 3: explore. Draw before you test: charts show odd values, skew and outliers that a single number hides.
- Count each category, sum up the numbers for each group, look at histograms and at `sdata.pairs`.

In the manual lesson 9, p. 48; lesson 12, p. 64.

Words to look up table `table.count_each()` p. 176; table `table.summary_of()` p. 179; table `table.correlation_between()` p. 182; chart `chart.histogram()` p. 115.

3 Try it yourself, without looking back

1. Make three summaries and two charts of your table that help to answer your question.

Worked answers: page 77, or `sdata.answers(23)`.

24

Day 24 of 30 • Python track

Project: figures

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Make the main figure of your project, ready for a journal. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word gives the average mark of each class? (day 1)
2. How do you make a normal distribution with average 50 and standard deviation 10? (day 19)
3. Which word keeps rows 10 to 20, by their place in the table? (day 6)
4. How do you make 100 random numbers with average 50 and standard deviation 10, the same every time? (day 9)

Answers: page 78, or `sdata.answers(24)`.

2 What to do

- Project day 4: figures. Make the one figure that answers your question best, with two panels (A and B), axis names with units, and a caption that says what the error bars or bands show.
- Use style `"paper"`, a size of 17 cm wide, and save it as `.pdf` and as a 600 dpi `.tif`.

In the manual lesson 7, p. 36; lesson 8, p. 41; lesson 15, p. 81.

Words to look up sdata `sdata.grid()` p. 149; chart `chart.style()` p. 142; chart `chart.size()` p. 142; chart `chart.save()` p. 112.

3 Try it yourself, without looking back

1. Make your main figure with two panels and save it ready for a journal.

Worked answers: page 78, or `sdata.answers(24)`.

25

Day 25 of 30 • Python track
Project: tests

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Choose and run the test that answers your question, with its effect size. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word gives the chance of a value above 80? (day 19)
2. How do you get the number in row 1, column 2 of a grid? (day 9)
3. Which word lists every step you did to a table? (day 6)
4. How do you give each class its own colour in a bar chart? (day 2)

Answers: page 78, or `sdata.answers(25)`.

2 What to do

- Project day 5: tests. Choose the test with the four questions of `"Which test when?"` before you run it, and run one test per question.
- Check the shape and the outliers first; use `based_on="ranks"` if the groups are small and far from bell-shaped.

In the manual lesson 14, p. 76; "Which test when?", p. 249.

Words to look up table `table.compare_averages()` p. 183; table `table.compare_counts()` p. 185; table `table.correlation_between()` p. 182; table `table.compare_shape_to()` p. 185.

3 Try it yourself, without looking back

1. Run the test that answers your question. Write down the test's name, the effect size and the p-value.

Worked answers: page 78, or `sdata.answers(25)`.

26

Day 26 of 30 • Python track

Project: models and fits

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Check your answer with a model of several columns, or a fitted line or formula. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word fits a distribution to a column? (day 19)
2. Which word draws one dot for each row, x against y? (day 3)
3. The same children were measured twice. Which word compares the two? (day 12)
4. Which word draws one small chart for each class? (day 5)

Answers: page 78, or `sdata.answers(26)`.

2 What to do

- Project day 6: go further. If several columns could explain the answer, put them in one model with `explain()`; if one column follows another, fit a line or a formula.
- Compare the model with the simple test of yesterday: does the answer hold when the other columns are taken into account?

In the manual lesson 16, p. 86; lesson 19, p. 99.

Words to look up table `table.explain()` p. 200; table `table.fit_line()` p. 188; table `table.add_predicted_column()` p. 200.

3 Try it yourself, without looking back

1. Explain your main column from two or three others, and say in one sentence what changed compared with the single test.

Worked answers: page 78, or `sdata.answers(26)`.

27

Day 27 of 30 • Python track

Project: results text

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Write the results, the methods and the figure caption of your project from the printed answers. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word hands a table on to pandas? (day 13)
2. Which word explains marks from hours, sleep and breakfast together? (day 15)
3. Which word tests whether the counts of categories differ between groups? (day 11)
4. Which word tests whether the average marks differ between the classes? (day 10)

Answers: page 79, or `sdata.answers(27)`.

2 What to do

- Project day 7: write. Turn the printed answers into a results paragraph: for each finding, the summary of each group with n, the effect size, the name of the test and the exact p-value.
- Write the methods from `table.steps()`: what you removed or filled, and which tests you used.
- Write the caption of your figure: what each panel shows, what the bars or bands are, and n.

In the manual "Writing it up", p. 250.
Words to look up table `table.steps()` p. 211.

3 Try it yourself, without looking back

1. Write a results paragraph, a methods paragraph and a figure caption for your project.

Worked answers: page 79, or `sdata.answers(27)`.

28

Day 28 of 30 • Python track

Review, and finish your project

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Remember words from the whole course, and run your project as one script from the raw file to the figure. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word counts the empty cells in each column? (day 8)
2. Which word draws survival curves? (day 17)
3. Which word adds the principal components PC1 and PC2? (day 16)
4. Which word estimates a column between the points you measured? (day 20)
5. Which word fits a straight line on a table? (day 18)
6. Which option makes a bar show the median instead of the average? (day 4)
7. Which word lists the practice tables? (day 1)
8. Which word gives the average of a column for each group? (day 9)
9. How do you look at the first 5 rows of a table? (day 6)
10. Which word moves the legend below the chart? (day 2)

Answers: page 79, or `sdata.answers(28)`.

2 What to do

- Answer the warm-up on paper, from memory, before you look anything up.
- For each question you could not answer, open its day with `sdata.day(n)` and run its example again.
- Project day 8: finish. Put every step of your project in one script, from the raw file to the figure, and run it from the top in a fresh session. It must run without errors.
- Read `steps()` out loud: would someone else understand what you did?

Words to look up table `table.steps()` p. 211; table `table.save()` p. 212.

3 Try it yourself, without looking back

1. Without looking back: which word fits a formula, which finds where it is biggest, and which gives the chance of a value above a number? Use one of them on any table or formula.
2. Run your whole project script from the top. Check that it makes the clean table, the figure and the numbers in your text.

Worked answers: page 79, or `sdata.answers(28)`.

29

Day 29 of 30 • Python track

Final test from memory

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Answer questions on the whole course and write a whole analysis from memory. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word finds where a formula is smallest? (day 19)
2. Which word compares one average with a number you choose? (day 12)
3. Which word puts two charts next to each other on one page? (day 5)
4. Which word draws a line with a dot at each point? (day 3)
5. How do you explain a yes-or-no column such as `"passed"`? (day 15)
6. Which word lists every change you made to a table, for your methods? (day 13)
7. Which word shows the rows with outliers? (day 11)
8. Which word says which pairs of groups differ? (day 10)
9. Which word fills an even map from scattered points, ready for a contour map? (day 20)
10. How do you fit a curve with x and x squared? (day 18)
11. Which word tests whether survival differs between groups? (day 17)
12. Which word puts rows that are alike into groups? (day 16)
13. Which word tells whether two columns go up and down together? (day 9)
14. How do you fill the empty marks with the median? (day 8)

Answers: page 80, or `sdata.answers(29)`.

2 What to do

- Today's warm-up is the final test: answer every question on paper, from memory. Then check your answers with `sdata.answers()`.
- Then write the whole program of the task in an empty file, without looking anything up. Only when it is finished, check it with `sdata.help`.

Words to look up sdata `sdata.words()` p. 238; sdata `sdata.help()` p. 238.

3 Try it yourself, without looking back

1. From memory, in an empty file: open the `"marks"` table, remove the rows with empty cells, give the average marks for each class, test the difference, and save a two-panel figure (boxes, and dots of hours against marks with a fitted line) as a PDF.

Worked answers: page 80, or `sdata.answers(29)`.

30

Day 30 of 30 • Python track

The same work in pandas and matplotlib

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. See what simplydata does underneath by writing the same analysis in pandas, SciPy and matplotlib. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word estimates a column between the points you measured? (day 20)
2. Which word finds where the two sides of an equation are equal? (day 19)
3. Which word fits a formula of your own, such as growth? (day 18)
4. Which word tests whether groups differ in their make-up across many columns (PERMANOVA)? (day 16)

Answers: page 81, or `sdata.answers(30)`.

2 What to do

- Today: the same work in pandas and matplotlib, the libraries simplydata uses underneath. Each word of the manual's dictionary says which one does the work (`"Underneath"`).
- Run the simplydata version and the pandas version of the task. Compare them line by line.
- Look back at the calendar with `sdata.day()` and choose three days to do again next month.

In the manual lesson 15, p. 81.

Words to look up table `table.as_pandas()` p. 212; table `table.as_polars()` p. 213; table `table.steps()` p. 211.

3 Try it yourself, without looking back

1. In simplydata: keep classes A and B of the marks table, give the average marks of each class, test the difference and draw it as bars.
2. Write the same analysis with pandas, SciPy and matplotlib, then compare the two line by line.

Worked answers: page 81, or `sdata.answers(30)`.

Beginner track

For people who have never programmed

Three days of just enough Python first, then about one lesson a day, with review days on days 7, 14 and 21, a small project of your own (days 27 and 28), a final test from memory and a look back. Tick each day when you have done it; the day's number is a link to its page.

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☐	30	Look back, and what comes next	-	66

1

Day 1 of 30 • Beginner track

Python first steps: values and names

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Run your first lines of Python: text in quotes, numbers, names for values, and print. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

Nothing to remember yet on day 1: just start.

2 What to do

- Open a place to type Python: a Jupyter notebook, VS Code, Spyder, or IDLE (it comes with Python). The manual's `"Install and first steps"` says how.
- Type each line of the example below, run it, and read what it prints.
- Then save the same lines in a file called day1.py and run the whole file: in a terminal, type python day1.py. A file of lines like this is called a script.

In the manual "Install and first steps", p. 4.

Example: type it yourself

```
name = "Asha"
age = 12
print(name)
print(age + 1)
print("Hello", name)
height_m = 1.44
print(height_m * 100)
```

What each line does

1 Give the text `"Asha"` the name name. Text always goes in quotes.
2 Give the number 12 the name age. Numbers have no quotes.
3 print shows a value on the screen: Asha.
4 Maths works as on a calculator: + - * / and ** for a power. This prints 13.
5 print can show several values, separated by commas.
6 A name has no spaces (use _ instead), and a decimal number has a point.
7 Prints 144.0: the height in centimetres.

3 Try it yourself, without looking back

1. Give the text `"Bergen"` the name city and the number 2250 the name rain (the rain in a year, in mm). Print the words `"Rain in"`, the city, and the rain of an average month (rain / 12).

Worked answers: page 82, or `sdata.answers(1)`.

2

Day 2 of 30 • Beginner track

Python first steps: lists and words

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Make lists, call a word with brackets and options, and use simplydata for the first time. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you write the text Asha in Python? (day 1)

2. How do you give the number 12 the name age? (day 1)

3. What does `print(age)` do? (day 1)

Answers: page 82, or `sdata.answers(2)`.

2 What to do

- Type and run the example below, one line at a time.
- Look closely at the brackets: square brackets [ ] make a list, round brackets ( ) call a word, and curly brackets { } make a table by hand.
- Then type `sdata.help("average")` and read what it says.

In the manual "Install and first steps", p. 4.

Words to look up sdata `sdata.numbers()` p. 214; nums `nums.average()` p. 226; nums `nums.keep_numbers()` p. 221.

Example: type it yourself

```
marks = [48, 71, 55, 88, 62]
print(len(marks))
print(round(3.14159, 2))
import simplydata as sdata
nums = sdata.numbers(marks)
nums.average()
nums.keep_numbers(above=50)
nums.show()
kids = {"name": ["Asha", "Ben"], "height": [144, 152]}
table = sdata.table(kids)
table.show()
```

What each line does

1. A list: values in square brackets, with commas between.
2. len is a word of Python that counts: 5. A word is called with round brackets around what it works on.
3. Two things in the brackets, separated by a comma: round 3.14159 to 2 decimals.
4. Load simplydata. From now on we call it by its short name, sdata.
5. numbers is a word of sdata: the point joins a thing and its word. We call the result nums.
6. A step on the numbers: it prints the answer in a sentence. Empty brackets still have to be there.
7. `above=50` is an option with a name: the name, =, the value.
8. Show the numbers that are left.
9. A table typed by hand: each column name with its list of values.
10. Turn it into a simplydata table.
11. Show the table.

3 Try it yourself, without looking back

1. Make a list of five temperatures, turn it into simplydata numbers, and find the biggest one and the median.

Worked answers: page 82, or `sdata.answers(2)`.

3

Day 3 of 30 • Beginner track

Reading an error, and your first chart

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Read an error message calmly from the bottom up, then make and save your first chart. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you write a list of the numbers 3, 5 and 2? (day 2)
2. In `nums.keep_numbers(above=50)`, what is `above=50`? (day 2)
3. How do you write the text Asha in Python? (day 1)

Answers: page 82, or `sdata.answers(3)`.

2 What to do

- Errors are normal: everybody gets them, every day. Today you make some on purpose.
- Type each of the errors to make on purpose, below, and read the error from the bottom up: the last line says what went wrong.
- Then start lesson 1, below.
- Work through `sdata.lesson(1)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(1, solutions=True)`.

In the manual lesson 1, p. 10; "Install and first steps", p. 4.

Words to look up sdata `sdata.help()` p. 238; sdata `sdata.example()` p. 111; sdata `sdata.chart()` p. 111; chart `chart.bars()` p. 113; chart `chart.save()` p. 112.

Errors to make on purpose (read the last line of each error first)

```
print(nmae)
```

NameError: a name is misspelt (nmae, not name), or was never given a value with = first.

```
print("Asha)
```

SyntaxError: a quote mark or a bracket is not closed. Every opening quote mark needs a closing one, and so does every bracket.

```
sdata.example("wether")
```

simplydata's own errors say what went wrong and suggest the word you probably meant.

```
nums = sdata.numbers([3, 5, 2])
nums.avrage()
```

A misspelt step: simplydata lists the steps you can use and the nearest one.

3 Try it yourself, without looking back

1. Find and fix the two mistakes in these lines, then run them: `nums = sdata.numbers([4, 8, 6)` and then `nums.averag()`
2. Make a bar chart of the `"weather"` table with `x="city"` and `y="rain"`, and save it as `"rain.png"`.

Worked answers: page 82, or `sdata.answers(3)`.

4

Day 4 of 30 • Beginner track
Titles and labels

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Put a title, axis names and a caption on a chart, and make its text easy to read. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Where in an error message do you look first? (day 3)
2. How do you give the number 12 the name age? (day 1)
3. What does NameError: name 'nmae' is not defined usually mean? (day 3)

Answers: page 83, or `sdata.answers(4)`.

2 What to do

- Work through `sdata.lesson(2)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(2, solutions=True)`.

In the manual lesson 2, p. 13.
Words to look up chart `chart.title_top()` p. 131; chart `chart.label_x()` p. 132; chart `chart.label_y()` p. 132; chart `chart.rotate_labels_x()` p. 138.

3 Try it yourself, without looking back

1. Draw dots of `"hours"` (x) against `"marks"` (y) from the marks table, with a title, a name for each axis and a caption at the bottom.

Worked answers: page 83, or `sdata.answers(4)`.

5 Colours, groups and legends

Day 5 of 30 • Beginner track

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Give each group its own colour and place the legend where it helps. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word puts a title above the chart? (day 4)
2. What does every simplydata program start with? (day 2)
3. How do you name the y-axis? (day 4)

Answers: page 83, or `sdata.answers(5)`.

2 What to do

- Work through `sdata.lesson(3)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(3, solutions=True)`.

In the manual lesson 3, p. 16.

Words to look up chart `chart.colour()` p. 140; chart `chart.colours()` p. 141; chart `chart.legend_bottom()` p. 144; chart `chart.order_x()` p. 146.

3 Try it yourself, without looking back

1. Weather table: draw lines of `"temperature"` over `"month"`, one colour for each city. Put the legend on top and use the style `"paper"`.

Worked answers: page 83, or `sdata.answers(5)`.

6

Day 6 of 30 • Beginner track
Different chart types

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Draw lines, dots, histograms, boxes, violins, pies and areas. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you give each class its own colour in a bar chart? (day 5)
2. Python says SyntaxError. What do you check first? (day 3)
3. Which word moves the legend below the chart? (day 5)

Answers: page 83, or `sdata.answers(6)`.

2 What to do

- Work through `sdata.lesson(4)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(4, solutions=True)`.

In the manual lesson 4, p. 21.
Words to look up chart `chart.lines()` p. 113; chart `chart.dots()` p. 114; chart `chart.histogram()` p. 115; chart `chart.boxes()` p. 116.

3 Try it yourself, without looking back

1. Draw violins of `"temperature"` for each city in the weather table, with all the dots on top.

Worked answers: page 83, or `sdata.answers(6)`.

7

Day 7 of 30 • Beginner track

Review of the first week

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Remember the Python and chart words of days 1 to 6 without looking, and use them together. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which chart shows how the values of one column are spread, in bins? (day 6)
2. The names under the bars bump into each other. Which word helps? (day 4)
3. Which word gives the average of numbers called nums? (day 2)
4. Which word lets you choose the colours of the groups? (day 5)
5. Which word lists the practice tables? (day 3)
6. What does `print(age)` do? (day 1)

Answers: page 84, or `sdata.answers(7)`.

2 What to do

- Answer the warm-up on paper, from memory, before you look anything up.
- For each question you could not answer, open its day with `sdata.day(n)` and run its example again.
- Then do the mixed task without looking back.

3 Try it yourself, without looking back

1. Make numbers of five marks and give their average. Then draw bars of the marks table with `x="subject"`, `y="marks"`, one colour for each class, and a title. Save it.

Worked answers: page 84, or `sdata.answers(7)`.

8

Day 8 of 30 • Beginner track

Averages, error bars and fit lines

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Show averages and medians with error bars, and draw a line through dots. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which two words show the spread of each group, with every value as a dot on top? (day 6)
2. How do you put the bars in your own order? (day 5)
3. How do you write the text Asha in Python? (day 1)

Answers: page 84, or `sdata.answers(8)`.

2 What to do

- Work through `sdata.lesson(5)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(5, solutions=True)`.

In the manual lesson 5, p. 26; "Standard deviation or standard error?", p. 244.

Words to look up option `show=` p. 121; option `error=` p. 121; chart `chart.numbers_on_bars()` p. 133; chart `chart.fit_line()` p. 122.

3 Try it yourself, without looking back

1. Weather table: draw bars of the average `"rain"` for each city with 95 % confidence intervals, and write the numbers on the bars.

Worked answers: page 84, or `sdata.answers(8)`.

9

Day 9 of 30 • Beginner track

Axes and reference lines

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Choose the range of an axis, and add lines and shading at the values that matter. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which option makes a bar show the median instead of the average? (day 8)
2. Which word draws one dot for each row, x against y? (day 6)
3. How do you write a list of the numbers 3, 5 and 2? (day 2)

Answers: page 85, or `sdata.answers(9)`.

2 What to do

- Work through `sdata.lesson(6)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(6, solutions=True)`.

In the manual lesson 6, p. 31.

Words to look up chart `chart.range_y()` p. 136; chart `chart.line_at_y()` p. 130; chart `chart.shade_between_y()` p. 131; chart `chart.log_scale_y()` p. 136.

3 Try it yourself, without looking back

1. Growth table: draw lines of `"population"` over `"year"`, one for each country, with a line at x = 2010 and the y-axis starting at 0.

Worked answers: page 85, or `sdata.answers(9)`.

10

Day 10 of 30 • Beginner track

Several charts together

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Put several charts on one page: one chart per group, side by side, or in a grid. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word sets the y-axis from 0 to 100? (day 9)
2. Which word draws a line with a dot at each point? (day 6)
3. How do you start a chart of a table called marks? (day 3)

Answers: page 85, or `sdata.answers(10)`.

2 What to do

- Work through `sdata.lesson(7)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(7, solutions=True)`.

In the manual lesson 7, p. 36.

Words to look up chart `chart.one_chart_per()` p. 148; sdata `sdata.side_by_side()` p. 148; sdata `sdata.one_above_other()` p. 149; sdata `sdata.grid()` p. 149.

3 Try it yourself, without looking back

1. Make a histogram and a box chart of the marks, and put them one above the other on one page.

Worked answers: page 85, or `sdata.answers(10)`.

11

Day 11 of 30 • Beginner track

A figure for a journal, and word families

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Size and save a figure for a journal, and learn the patterns that make words easy to guess. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word draws one small chart for each class? (day 10)
2. Which option adds error bars that say how sure we are of the average? (day 8)
3. Which word makes all the text on a chart bigger or smaller? (day 4)

Answers: page 85, or `sdata.answers(11)`.

2 What to do

- Work through `sdata.lesson(8)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(8, solutions=True)`.
- Read `"How the words are built"` in the manual (two pages). It is the key to guessing words you have never seen.
- Type `sdata.words()` and look for the families: `keep_` and `remove_`, `..._x` and `..._y`, `add_..._column`, `..._of`.
- Then do the task: guess first, check with `sdata.help` after.

In the manual lesson 8, p. 41; "How the words are built", p. 5.

Words to look up chart `chart.second_y()` p. 139; chart `chart.size()` p. 142; chart `chart.style()` p. 142; chart `chart.note()` p. 133; chart `chart.save()` p. 112; chart `chart.hide_legend()` p. 145; chart `chart.legend_left()` p. 144; sdata `sdata.words()` p. 238.

3 Try it yourself, without looking back

1. A figure for a journal: dots of `"hours"` against `"marks"`, style `"paper"`, 8.5 cm wide and 7 cm high, font size 8, saved as a `.tif` with 600 dpi.
2. Save one chart as `.png`, `.pdf` and `.svg`. Then guess, before you look anything up: which word hides the legend, and which puts it on the left? Try your guesses on a chart with `colour=`.

Worked answers: page 85, or `sdata.answers(11)`.

12

Day 12 of 30 • Beginner track

Open a table and look at it

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Open a table, look at its first rows and columns, and count its rows. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word adds a second y-axis on the right? (day 11)
2. Which word draws a line across at y = 50, with a label? (day 9)
3. How do you give each class its own colour in a bar chart? (day 5)

Answers: page 86, or `sdata.answers(12)`.

2 What to do

- Work through `sdata.lesson(9)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(9, solutions=True)`.

In the manual lesson 9, p. 48.

Words to look up sdata `sdata.table()` p. 151; table `table.show_first_rows()` p. 152; table `table.describe_columns()` p. 153; table `table.count_rows()` p. 176.

3 Try it yourself, without looking back

1. Open the `"shop"` practice table. How many rows does it have, and how many rows does each shop have?

Worked answers: page 86, or `sdata.answers(12)`.

13

Day 13 of 30 • Beginner track

Choose rows and columns, and undo

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Keep and remove rows and columns, sort, and take back any step. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you look at the first 5 rows of a table? (day 12)
2. Which word puts two charts next to each other on one page? (day 10)
3. Which chart shows how the values of one column are spread, in bins? (day 6)

Answers: page 86, or `sdata.answers(13)`.

2 What to do

- Work through `sdata.lesson(10)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(10, solutions=True)`.

In the manual lesson 10, p. 52.

Words to look up table `table.keep_rows()` p. 157; table `table.keep_columns()` p. 155; table `table.sort_by()` p. 169; table `table.undo()` p. 211.

3 Try it yourself, without looking back

1. Weather table: keep the city `"Sunnyvale"`, keep only the columns month, temperature and rain, sort by rain with the biggest first, and save it as `"sunny.csv"`.

Worked answers: page 86, or `sdata.answers(13)`.

14

Day 14 of 30 • Beginner track

Review of the second week

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Remember the words of days 8 to 13 without looking, and use them together. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word keeps only the rows where subject is Maths? (day 13)
2. Which word draws one bar for each name, as tall as the marks? (day 3)
3. Which word draws a straight line through the dots? (day 8)
4. Which word sets the size of a figure in cm? (day 11)
5. A few values are much bigger than the rest. Which word helps? (day 9)
6. Which word describes each column: its kind, empty cells and numbers? (day 12)
7. How do you put four charts in two rows, with the letters A to D? (day 10)
8. How do you give the number 12 the name age? (day 1)

Answers: page 87, or `sdata.answers(14)`.

2 What to do

- Answer the warm-up on paper, from memory, before you look anything up.
- For each question you could not answer, open its day with `sdata.day(n)` and run its example again.
- Then do the mixed task without looking back.

3 Try it yourself, without looking back

1. Weather table: keep the city `"Frostby"`, sort by `"temperature"`, and draw a line of the temperature over the month, with a line at y = 0 and the label `"freezing"`.

Worked answers: page 87, or `sdata.answers(14)`.

15

Day 15 of 30 • Beginner track

Clean the data and add columns

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Find and fill empty cells, fix text and numbers, and add new columns. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word sorts with the biggest marks first? (day 13)
2. Which word counts the rows of each fruit? (day 12)
3. Which word checks by eye whether values are bell-shaped? (day 8)
4. What does `print(age)` do? (day 1)

Answers: page 87, or `sdata.answers(15)`.

2 What to do

- Work through `sdata.lesson(11)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(11, solutions=True)`.

In the manual lesson 11, p. 57; "Missing values and imputation", p. 248.

Words to look up table `table.count_empty_cells()` p. 163; table `table.fill_empty_cells()` p. 164; table `table.make_numbers()` p. 165; table `table.add_column()` p. 170.

3 Try it yourself, without looking back

1. Shop table: count the empty cells, fill the empty `"sales"` with the line between neighbours for each shop (sorted by `"date"`), and add a column `"weekday"` with the name of the weekday.

Worked answers: page 87, or `sdata.answers(15)`.

16

Day 16 of 30 • Beginner track

Summaries, and from a table to a chart

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Sum up columns for each group, and draw a summary as a chart. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word counts the empty cells in each column? (day 15)
2. You made a mistake in the last step. Which word takes it back? (day 13)
3. Which word shows the numbers on the y-axis as percents? (day 9)
4. In `nums.keep_numbers(above=50)`, what is `above=50`? (day 2)

Answers: page 88, or `sdata.answers(16)`.

2 What to do

- Work through `sdata.lesson(12)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(12, solutions=True)`.

In the manual lesson 12, p. 64.

Words to look up table `table.average_of()` p. 177; table `table.summary_of()` p. 179; table `table.correlation_between()` p. 182; table `table.join_with()` p. 209.

3 Try it yourself, without looking back

1. Fruit table: the total `"count"` of each fruit, drawn as bars with the numbers on the bars.

Worked answers: page 88, or `sdata.answers(16)`.

17

Day 17 of 30 • Beginner track

Lists and grids of numbers

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Do maths on lists and grids of numbers, and make random numbers. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word gives the average of a column for each group? (day 16)
2. Which word keeps rows 10 to 20, by their place in the table? (day 13)
3. Which word draws one small chart for each class? (day 10)
4. How do you save a chart as a PDF? (day 3)

Answers: page 88, or `sdata.answers(17)`.

2 What to do

- Work through `sdata.lesson(13)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(13, solutions=True)`.

In the manual lesson 13, p. 72.

Words to look up sdata `sdata.numbers()` p. 214; sdata `sdata.random_numbers()` p. 214; nums `nums.median()` p. 226; nums `nums.percentile()` p. 228.

3 Try it yourself, without looking back

1. Make 200 random heights with average 150 and standard deviation 8 (`seed=1`). What is their 90th percentile? Draw them as a histogram.

Worked answers: page 88, or `sdata.answers(17)`.

18

Day 18 of 30 • Beginner track

Is a difference real? Tests and effect sizes

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Test whether averages differ, and say how big the difference is. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you make simplydata numbers from a list? (day 17)
2. How do you fill the empty marks with the median? (day 15)
3. How do you save a picture sharp enough for a journal? (day 11)
4. Which word puts a title above the chart? (day 4)

Answers: page 89, or `sdata.answers(18)`.

2 What to do

- Work through `sdata.lesson(14)`, examples 1 and 3: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(14, solutions=True)`.
- Read `"The null hypothesis and the p-value"` and `"Effect sizes"` in the manual.

In the manual lesson 14, p. 76; "The null hypothesis and the p-value", p. 245; "Effect sizes: how big is the difference?", p. 246.

Words to look up table `table.compare_averages()` p. 183; table `table.compare_each_pair()` p. 183; table `table.fit_line()` p. 188.

3 Try it yourself, without looking back

1. School table: do the average marks differ between pupils who had breakfast and pupils who did not? Write down the difference, the effect size and the p-value.

Worked answers: page 89, or `sdata.answers(18)`.

19

Day 19 of 30 • Beginner track

Counts, outliers and many tests

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Test counts of categories, and check shapes and outliers. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word tests whether the average marks differ between the classes? (day 18)
2. Which word tells whether two columns go up and down together? (day 16)
3. Which word counts the rows? (day 12)
4. Which word moves the legend below the chart? (day 5)

Answers: page 89, or `sdata.answers(19)`.

2 What to do

- Work through `sdata.lesson(14)`, examples 2, 4 and 5: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(14, solutions=True)`.
- Read `"Many tests at once"` in the manual.

In the manual lesson 14, p. 76; "Many tests at once", p. 246.

Words to look up table `table.compare_counts()` p. 185; table `table.show_rows_with_outliers()` p. 162; table `table.compare_shape_to()` p. 185; table `table.add_adjusted_p_value_column()` p. 186.

3 Try it yourself, without looking back

1. School table: is the share of pupils who `"passed"` different with and without breakfast?

Worked answers: page 89, or `sdata.answers(19)`.

20

Day 20 of 30 • Beginner track

Choosing a test, and a whole analysis

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Choose the right test, and go from a question to a tested answer and a figure. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word tests whether the counts of categories differ between groups? (day 19)
2. Which word gives the middle value of numbers? (day 17)
3. Which word lists every step you did to a table? (day 13)
4. Which two words show the spread of each group, with every value as a dot on top? (day 6)

Answers: page 89, or `sdata.answers(20)`.

2 What to do

- Read `"Which test when?"` and `"Writing it up"` in the manual. Answer its four questions for each task below before you run anything.
- Run `sdata.help("compare_before_after")` and `sdata.help("compare_average_to")` and type their examples.
- Work through `sdata.lesson(15)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(15, solutions=True)`.

In the manual lesson 15, p. 81; "Which test when?", p. 249; "Writing it up", p. 250.

Words to look up table `table.compare_average_to()` p. 184; table `table.compare_before_after()` p. 184; table `table.compare_percent_to()` p. 187; table `table.steps()` p. 211; sdata `sdata.grid()` p. 149; table `table.save()` p. 212; table `table.as_pandas()` p. 212.

3 Try it yourself, without looking back

1. Marks table: is each child's Maths mark different from the same child's Science mark? Choose the right word first.
2. School table: is the average `"sleep"` different from 8 hours?
3. From memory: weather table, compare the average `"temperature"` of the cities, draw boxes with all the dots, and save the figure as a PDF and the table as `"weather.xlsx"`.

Worked answers: page 89, or `sdata.answers(20)`.

21 Review of the third week

Day 21 of 30 • Beginner track

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Remember the words of days 15 to 20 without looking, and use them together. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. The same children were measured twice. Which word compares the two? (day 20)
2. Which word turns a column of text such as `"144"` into numbers? (day 15)
3. Which word writes a note with an arrow on the chart? (day 11)
4. Which word says which pairs of groups differ? (day 18)
5. Which word joins two tables on a column they share? (day 16)
6. Which word shows the rows with outliers? (day 19)
7. How do you make 100 random numbers with average 50 and standard deviation 10, the same every time? (day 17)
8. Which option makes a bar show the median instead of the average? (day 8)
9. How do you write the text Asha in Python? (day 1)

Answers: page 90, or `sdata.answers(21)`.

2 What to do

- Answer the warm-up on paper, from memory, before you look anything up.
- For each question you could not answer, open its day with `sdata.day(n)` and run its example again.
- Then do the mixed task without looking back.

3 Try it yourself, without looking back

1. School table: the average `"marks"` for each value of `"breakfast"`, a test of the difference, and boxes of the marks for each group.

Worked answers: page 90, or `sdata.answers(21)`.

22

Day 22 of 30 • Beginner track

Models: explain one column from several

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Explain one column from several others with a model. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word compares one average with a number you choose? (day 20)
2. Which word checks whether values are bell-shaped (normal)? (day 19)
3. Which word adds a new column made from other columns? (day 15)
4. Which option adds error bars that say how sure we are of the average? (day 8)

Answers: page 91, or `sdata.answers(22)`.

2 What to do

- Work through `sdata.lesson(16)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(16, solutions=True)`.

In the manual lesson 16, p. 86.

Words to look up table `table.explain()` p. 200; table `table.add_predicted_column()` p. 200.

3 Try it yourself, without looking back

1. School table: explain `"days_absent"` using `"sleep"` and `"breakfast"` with `outcome="counts"`. Does breakfast matter?

Worked answers: page 91, or `sdata.answers(22)`.

23

Day 23 of 30 • Beginner track

Patterns across many columns

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Find patterns across many columns with principal components and clusters. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word explains marks from hours, sleep and breakfast together? (day 22)
2. What should a results sentence give? (day 20)
3. Which word gives several summaries at once? (day 16)
4. Which word sets the y-axis from 0 to 100? (day 9)

Answers: page 91, or `sdata.answers(23)`.

2 What to do

- Work through `sdata.lesson(17)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(17, solutions=True)`.

In the manual lesson 17, p. 90.

Words to look up table `table.add_principal_components()` p. 201; table `table.add_clusters()` p. 203; table `table.add_nmds()` p. 204; table `table.compare_profiles()` p. 205.

3 Try it yourself, without looking back

1. Meadow table: add 3 clusters and count the rows of each cluster across `"field"`.

Worked answers: page 91, or `sdata.answers(23)`.

24

Day 24 of 30 • Beginner track

Survival, and planning a study

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Draw survival curves, and work out how many values a study needs. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word adds the principal components PC1 and PC2? (day 23)
2. Which word lists every change you made to a table, for your methods? (day 20)
3. How do you get the number in row 1, column 2 of a grid? (day 17)
4. Which word puts two charts next to each other on one page? (day 10)

Answers: page 91, or `sdata.answers(24)`.

2 What to do

- Work through `sdata.lesson(18)`: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(18, solutions=True)`.

In the manual lesson 18, p. 95; "Two kinds of mistake, and power", p. 246.

Words to look up table `table.survival_over_time()` p. 205; chart `chart.survival_curves()` p. 207; table `table.compare_survival()` p. 206; sdata `sdata.how_many_needed()` p. 207.

3 Try it yourself, without looking back

1. How many values does each group need to find a large difference (`effect_size=0.8`) with power 0.8? And with a small one (0.2)?

Worked answers: page 91, or `sdata.answers(24)`.

25

Day 25 of 30 • Beginner track

Fitting, finding and solving

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Fit lines, curves and formulas, find where a formula is biggest, and solve equations. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word draws survival curves? (day 24)
2. How do you explain a yes-or-no column such as `"passed"`? (day 22)
3. A p-value answers `"could it be chance?"`. What answers `"how big is it?"` (day 18)
4. If `keep_rows` keeps rows, what would you guess takes columns away? (day 11)

Answers: page 92, or `sdata.answers(25)`.

2 What to do

- Work through `sdata.lesson(19)`, examples 1 and 2: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(19, solutions=True)`.
- Work through `sdata.lesson(19)`, example 3: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(19, solutions=True)`.

In the manual lesson 19, p. 99; "Fitting a line or a formula", p. 247.

Words to look up table `table.fit_line()` p. 188; table `table.fit_curve()` p. 189; table `table.fit_formula()` p. 189; chart `chart.fit_formula()` p. 123; sdata `sdata.find_smallest()` p. 190; sdata `sdata.find_biggest()` p. 191; sdata `sdata.solve()` p. 191.

3 Try it yourself, without looking back

1. Growth table: fit the formula `"forest = a + b * (year - 2000)"` for each country. Which country loses forest fastest?
2. A box has a square bottom of side x and a height of 10 - x, so its volume is x * x * (10 - x). For which x between 0 and 10 is the volume biggest?

Worked answers: page 92, or `sdata.answers(25)`.

26

Day 26 of 30 • Beginner track

Distributions and values between points

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Work out chances from a distribution, and estimate values between measured points. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word fits a straight line on a table? (day 25)
2. Which word puts rows that are alike into groups? (day 23)
3. You did many tests at once. Which word corrects the p-values? (day 19)
4. How do you look at the first 5 rows of a table? (day 12)

Answers: page 92, or `sdata.answers(26)`.

2 What to do

- Work through `sdata.lesson(20)`, examples 1 and 2: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(20, solutions=True)`.
- Work through `sdata.lesson(20)`, example 3: type each example yourself (do not copy and paste), run it and read what each line does. Then do the lesson's tasks and check them with `sdata.lesson(20, solutions=True)`.

In the manual lesson 20, p. 104; "The shape of the values: distributions", p. 244.

Words to look up sdata `sdata.distribution()` p. 193; dist `dist.chance()` p. 195; dist `dist.percentile()` p. 196; table `table.fit_distribution()` p. 194; table `table.values_at()` p. 192; table `table.map_of()` p. 193.

3 Try it yourself, without looking back

1. Heights are normal with average 150 and standard deviation 8. What share is taller than 165? Which height do 95 % of them stay below?
2. Weather table: estimate the `"rain"` of each city at `month_number` 2.5 and 7.5, joined by straight lines.

Worked answers: page 92, or `sdata.answers(26)`.

27

Day 27 of 30 • Beginner track

Small project: look, clean and sum up

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Choose a question and a file, then look at it, clean it and sum it up. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. How do you make a normal distribution with average 50 and standard deviation 10? (day 26)
2. Which word cuts a column of numbers into named groups? (day 15)
3. How do you give the panels of a figure the letters A and B? (day 20)
4. Which word tests whether survival differs between groups? (day 24)

Answers: page 93, or `sdata.answers(27)`.

2 What to do

- Your small project, day 1. Choose data of your own (a file from school, work or a public data set), or use the `"school"` practice table.
- Write your question in one sentence. Open the file, look at it, clean what needs cleaning and make two summaries.

In the manual lesson 9, p. 48; lesson 11, p. 57; lesson 12, p. 64.

Words to look up sdata `sdata.table()` p. 151; table `table.describe_columns()` p. 153; table `table.count_empty_cells()` p. 163; table `table.summary_of()` p. 179.

3 Try it yourself, without looking back

1. Write your question. Open your file, describe its columns, deal with the empty cells and make two summaries for each group.

Worked answers: page 93, or `sdata.answers(27)`.

28

Day 28 of 30 • Beginner track

Small project: test, figure and results

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Answer your question with a test, a figure and two sentences of results. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word adds the values a model predicts as a new column? (day 22)
2. Which option gives the rank-based test, for small groups that are not bell-shaped? (day 18)
3. A word for one axis ends in ...? (day 11)
4. How do you fit a curve with x and x squared? (day 25)

Answers: page 93, or `sdata.answers(28)`.

2 What to do

- Your small project, day 2. Test the difference, make one clear figure, and write two sentences with the numbers as they were printed.

In the manual lesson 14, p. 76; lesson 15, p. 81; "Writing it up", p. 250.
Words to look up table `table.compare_averages()` p. 183; chart `chart.boxes()` p. 116; chart `chart.save()` p. 112.

3 Try it yourself, without looking back

1. Test your question, draw one figure and save it, and write two sentences of results.

Worked answers: page 93, or `sdata.answers(28)`.

29

Day 29 of 30 • Beginner track

Final test from memory

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Answer questions on the whole course and write a whole analysis from memory. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word tests whether groups differ in their make-up across many columns (PERMANOVA)? (day 23)
2. Which word gives the chance of a value above 80? (day 26)
3. Which word says how many values you need in each group? (day 24)
4. Which word hands a table on to pandas? (day 20)
5. Which word fits a formula of your own, such as growth? (day 25)
6. Which file endings give a picture that stays sharp at any size? (day 11)
7. What does every simplydata program start with? (day 2)
8. Where in an error message do you look first? (day 3)
9. How do you name the y-axis? (day 4)
10. Which word lets you choose the colours of the groups? (day 5)
11. Which word keeps only the rows where subject is Maths? (day 13)
12. Which word draws one dot for each row, x against y? (day 6)
13. How do you give the number 12 the name age? (day 1)
14. Which word draws a straight line through the dots? (day 8)

Answers: page 94, or `sdata.answers(29)`.

2 What to do

- Today's warm-up is the final test: answer every question on paper, from memory. Then check your answers with `sdata.answers()`.
- Then write the whole program of the task in an empty file, without looking anything up. Only when it is finished, check it with `sdata.help`.

Words to look up sdata `sdata.words()` p. 238; sdata `sdata.help()` p. 238.

3 Try it yourself, without looking back

1. From memory, in an empty file: open the `"marks"` table, remove the rows with empty cells, give the average marks for each class, test the difference, and save a two-panel figure (boxes, and dots of hours against marks with a fitted line) as a PDF.

Worked answers: page 94, or `sdata.answers(29)`.

30

Day 30 of 30 • Beginner track

Look back, and what comes next

☐ Warm-up ☐ What to do ☐ Task Date: _____

Goal. Do your hardest day again, and take a first look at pandas underneath simplydata. About an hour.

1 Warm-up, 5 minutes: from memory, without looking back

1. Which word fits a distribution to a column? (day 26)
2. Which word finds where a formula is smallest? (day 25)
3. Which word adds the principal components PC1 and PC2? (day 23)
4. Which word gives the average of a column for each group? (day 16)

Answers: page 94, or `sdata.answers(30)`.

2 What to do

- Look back over the 30 days with `sdata.day()`: it lists every day.
- Choose the three days you found hardest and do their tasks again, without looking back.
- Take a first look at pandas, the library underneath simplydata tables: `table.as_pandas()` hands your table over, and the manual's dictionary says which library does the work of each word.

Words to look up sdata `sdata.day()` p. 240; sdata `sdata.words()` p. 238; table `table.as_pandas()` p. 212.

3 Try it yourself, without looking back

1. Do the task of your hardest day again from memory. Then hand the marks table to pandas and print its first rows with pandas' own word `head()`.

Worked answers: page 94, or `sdata.answers(30)`.

Answers: Python track

The warm-up answers and a worked answer to each task

There is often more than one good answer: if yours runs and answers the question, it is right. Every piece of code here runs as it is (the tests run it).

Day 1: Your first chart and your first table

Task 1. Make a bar chart of the `"weather"` table with `x="city"` and `y="rain"`, and save it as `"rain.png"`.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.bars(x="city", y="rain")
chart.save("rain.png")
```

> One bar for each city, as tall as its average rain over the 12 months.

Task 2. Draw dots of `"hours"` (x) against `"marks"` (y) from the marks table, with a title, a name for each axis and a caption at the bottom.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.title_top("More study, better marks?")
chart.label_x("Hours of study")
chart.label_y("Mark (out of 100)")
chart.caption_bottom("Made-up data")
chart.save("hours_and_marks.png")
```

> First draw, then add the words; the order of the word steps does not matter.

Task 3. Start a table of the weather, keep the rows with `"temperature"` above 15, and give the average `"rain"` for each city.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.keep_rows("temperature", above=15)
table.average_of("rain", for_each="city")
```

> `keep_rows` says how many warm months are left; `average_of` gives one average for each city.

Back to day 1

Day 2: Colours, word families and saving

1. Which word lists the practice tables? (day 1)
   → `sdata.example()`; `sdata.example("marks")` gets one of them.
2. How do you start a chart of a table called marks? (day 1)
   → chart = `sdata.chart(marks)`
3. Which word draws one bar for each name, as tall as the marks? (day 1)
   → `chart.bars(x="name", y="marks")`

Task 1. Weather table: draw lines of `"temperature"` over `"month"`, one colour for each city. Put the legend on top and use the style `"paper"`.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city")
chart.legend_top()
chart.style("paper")
```

```
chart.save("temperature.png")
```

`colour="city"` draws one line for each city and adds a legend; the style changes the whole look.

Task 2. Save one chart as `.png`, `.pdf` and `.svg`. Then guess, before you look anything up: which word hides the legend, and which puts it on the left? Try your guesses on a chart with `colour=`.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks", colour="class")
chart.save("subjects.png")
chart.save("subjects.pdf")
chart.save("subjects.svg")
chart.legend_left()
chart.hide_legend()
chart.save("no_legend.png")
```

`legend_left` follows `legend_bottom` and `legend_top`; `hide_legend` follows `show_` and `hide_`. A family is learned once.

Back to day 2

Day 3: Every kind of chart

1. How do you give each class its own colour in a bar chart? (day 2)
   → `colour="class"` inside the chart step: `chart.bars(x="subject", y="marks", colour="class")`
2. Which word moves the legend below the chart? (day 2)
   → `chart.legend_bottom()`
3. How do you save a chart as a PDF? (day 1)
   → `chart.save("my_chart.pdf")`: the end of the file name decides the type.

Task 1. Draw violins of `"temperature"` for each city in the weather table, with all the dots on top.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.violins(x="city", y="temperature")
chart.all_dots(x="city", y="temperature")
chart.save("violins.png")
```

A violin shows the shape of the spread; the dots show every month.

Back to day 3

Day 4: Statistics on charts, and axes

1. Which chart shows how the values of one column are spread, in bins? (day 3)
   → `chart.histogram(x="marks", bins=8)`
2. Which word puts a title above the chart? (day 1)
   → `chart.title_top("My title")`
3. Which two words show the spread of each group, with every value as a dot on top? (day 3)
   → `chart.boxes(x="class", y="marks")` and `chart.all_dots(x="class", y="marks")`

Task 1. Weather table: draw bars of the average `"rain"` for each city with 95 % confidence intervals, and write the numbers on the bars.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.bars(x="city", y="rain", error="confidence interval")
chart.numbers_on_bars()
chart.save("rain_ci.png")
```

A line is printed saying which error bars were drawn, so the caption can say it too.

Task 2. Growth table: draw lines of `"population"` over `"year"`, one for each country, with a line at x = 2010 and the y-axis starting at 0.

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.lines(x="year", y="population", colour="country")
chart.line_at_x(2010, label="2010")
chart.range_y(0, 60)
chart.save("growth.png")
```

`range_y` takes the lowest and the highest value of the axis; `line_at_x` draws a line going up.

Back to day 4

Day 5: Several charts, and a figure for a journal

1. Which option makes a bar show the median instead of the average? (day 4)
   → `show="median"`
2. Which word lets you choose the colours of the groups? (day 2)
   → `chart.colours("orange", "teal", "purple")`
3. Which option adds error bars that say how sure we are of the average? (day 4)
   → `error="standard error"`, or `error="confidence interval"` for the 95 % interval.

Task 1. Make a histogram and a box chart of the marks, and put them one above the other on one page.

```
import simplydata as sdata
marks = sdata.example("marks")
a = sdata.chart(marks).histogram(x="marks")
b = sdata.chart(marks).boxes(x="class", y="marks")
page = sdata.one_above_other(a, b)
page.save("two_charts.png")
```

Each chart is made first; `one_above_other` puts them on one page.

Task 2. A figure for a journal: dots of `"hours"` against `"marks"`, style `"paper"`, 8.5 cm wide and 7 cm high, font size 8, saved as a `.tif` with 600 dpi.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.style("paper")
chart.size(width=8.5, height=7, unit="cm")
chart.font_size(8)
chart.save("figure1.tif", resolution=600)
```

Journals often ask for 8.5 cm (one column) or 17 cm (two columns) and 300 to 600 dpi.

Back to day 5

Day 6: Open a table, choose rows and columns

1. Which word draws one small chart for each class? (day 5)
   → `chart.one_chart_per("class")`
2. Which word draws one dot for each row, x against y? (day 3)
   → `chart.dots(x="hours", y="marks")`
3. Which word puts two charts next to each other on one page? (day 5)
   → `sdata.side_by_side(a, b)`

Task 1. Open the `"shop"` practice table. How many rows does it have, and how many rows does each shop have?

```python
import simplydata as sdata
table = sdata.table(sdata.example("shop"))
table.count_rows()
table.count_each("shop")
table.describe_columns()
```

`count_rows` counts all the rows, `count_each` the rows of each shop; `describe_columns` also shows the empty cells.

Task 2. Weather table: keep the city `"Sunnyvale"`, keep only the columns month, temperature and rain, sort by rain with the biggest first, and save it as `"sunny.csv"`.

```python
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.keep_rows("city", equal_to="Sunnyvale")
table.keep_columns("month", "temperature", "rain")
table.sort_by("rain", biggest_first=True)
table.save("sunny.csv")
table.steps()
```

Each step prints what it did; `steps()` lists them all, in order.

Back to day 6

Day 7: Review of the first week

1. How do you look at the first 5 rows of a table? (day 6)
   → `table.show_first_rows(5)`
2. How do you name the y-axis? (day 1)
   → `chart.label_y("Marks out of 100")`
3. Which word draws a straight line through the dots? (day 4)
   → `chart.fit_line()`, after `chart.dots(...)`
4. How do you put the bars in your own order? (day 2)
   → `chart.order_x("C", "B", "A")`
5. How do you put four charts in two rows, with the letters A to D? (day 5)
   → `sdata.grid(a, b, c, d, per_row=2, letters=True)`
6. Which word draws a line with a dot at each point? (day 3)
   → `chart.lines(x="month", y="temperature", dots=True)`

Task 1. Marks table: keep only class `"A"`, draw bars of the average `"marks"` for each subject with standard error bars, give the chart a title and save it as a PDF.

```python
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("class", equal_to="A")
chart = table.chart()
chart.bars(x="subject", y="marks", error="standard error")
chart.title_top("Class A")
chart.save("class_a.pdf")
```

Day 1 (bars, save), day 4 (`error=`) and day 6 (`keep_rows`) in one small program.

Back to day 7

Day 8: Clean the data and add columns

1. Which word describes each column: its kind, empty cells and numbers? (day 6)
   → `table.describe_columns()`
2. Which word adds a second y-axis on the right? (day 5)
   → `chart.second_y("rain")`
3. The names under the bars bump into each other. Which word helps? (day 1)
   → `chart.rotate_labels_x(45)`

Task 1. Shop table: count the empty cells, fill the empty `"sales"` with the line between neighbours for each shop (sorted by `"date"`), and add a column `"weekday"` with the name of the weekday.

```
import simplydata as sdata
table = sdata.table(sdata.example("shop"))
table.count_empty_cells()
table.fill_empty_cells("sales", with_the="line between neighbours", sorted_by="date", for_each="shop")
table.add_date_part_column("weekday", from_column="date", part="weekday name")
table.show_first_rows()
```

`for_each=` keeps the two shops apart, so a gap is never filled from the other shop's days.

Back to day 8

Day 9: Summaries, numbers and grids

1. Which word counts the empty cells in each column? (day 8)
   → `table.count_empty_cells()`
2. Which word counts the rows of each fruit? (day 6)
   → `table.count_each("fruit")`
3. If `keep_rows` keeps rows, what would you guess takes columns away? (day 2)
   → `remove_columns`: verb + thing. `keep_` and `remove_` work on rows, columns and numbers.

Task 1. Fruit table: the total `"count"` of each fruit, drawn as bars with the numbers on the bars.

```
import simplydata as sdata
table = sdata.table(sdata.example("fruit"))
totals = table.total_of("count", for_each="fruit")
chart = totals.chart()
chart.bars(x="fruit", y="total of count")
chart.numbers_on_bars()
chart.save("fruit_totals.png")
```

A summary gives back a new table, with a column called `"total of count"`, ready for a chart.

Task 2. Make 200 random heights with average 150 and standard deviation 8 (`seed=1`). What is their 90th percentile? Draw them as a histogram.

```
import simplydata as sdata
heights = sdata.random_numbers(how_many=200, average=150, standard_deviation=8, seed=1)
heights.percentile(90)
chart = heights.chart()
chart.histogram(x="value")
chart.save("heights.png")
```

About 157: nine heights in ten are below it. `seed=1` gives the same numbers every time.

Back to day 9

Day 10: Is a difference real? Tests and effect sizes

1. Which word gives the average of a column for each group? (day 9)
   → `table.average_of("marks", for_each="class")`
2. Which word counts the rows? (day 6)
   → `table.count_rows()`
3. Which chart shows how the values of one column are spread, in bins? (day 3)
   → `chart.histogram(x="marks", bins=8)`

Task 1. School table: do the average marks differ between pupils who had breakfast and pupils who did not? Write down the difference, the effect size and the p-value.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.average_of("marks", for_each="breakfast")
table.compare_averages("marks", for_each="breakfast")
```

Two groups, so it is Welch's t-test; the sentence gives the difference, Hedges' g and the p-value.

Back to day 10

Day 11: Counts, outliers and many tests

1. Which word tests whether the average marks differ between the classes? (day 10)
   → `table.compare_averages("marks", for_each="class")`
2. How do you fill the empty marks with the median? (day 8)
   → `table.fill_empty_cells("marks", with_the="median")`
3. Which word checks by eye whether values are bell-shaped? (day 4)
   → `chart.qq_dots("marks")`

Task 1. School table: is the share of pupils who `"passed"` different with and without breakfast?

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.count_rows(for_each="breakfast", across="passed")
table.compare_counts("passed", for_each="breakfast")
```

passed is a category (yes or no), so the test compares counts: the chi-square test, with Cramer's V as the effect size.

Back to day 11

Day 12: Choosing a test, and writing it up

1. Which word tests whether the counts of categories differ between groups? (day 11)
   → `table.compare_counts("result", for_each="class")`
2. Which word tells whether two columns go up and down together? (day 9)
   → `table.correlation_between("hours", "marks")`
3. Which word sets the size of a figure in cm? (day 5)
   → `chart.size(width=17, height=8, unit="cm")`

Task 1. Marks table: is each child's Maths mark different from the same child's Science mark? Choose the right word first.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_columns("name", "subject", "marks")
table.turn_rows_into_columns(names_from="subject", values_from="marks")
table.compare_before_after("Maths", "Science")
```

The same children twice, so the test must be paired. First one row per child, with a Maths and a Science column.

Task 2. School table: is the average `"sleep"` different from 8 hours?

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.compare_average_to("sleep", 8)
```

One average against a number you choose: the one-sample t-test. Empty sleep cells are skipped, with a note.

Back to day 12

Day 13: A whole analysis

1. The same children were measured twice. Which word compares the two? (day 12)
   → `table.compare_before_after("before", "after")`: the paired test.
2. Which word says which pairs of groups differ? (day 10)
   → `table.compare_each_pair("marks", for_each="class")`

3. Which word keeps only the rows where subject is Maths? (day 6)
   → `table.keep_rows("subject", equal_to="Maths")`

Task 1. From memory: weather table, compare the average `"temperature"` of the cities, draw boxes with all the dots, and save the figure as a PDF and the table as `"weather.xlsx"`.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.compare_averages("temperature", for_each="city")
chart = table.chart()
chart.boxes(x="city", y="temperature")
chart.all_dots(x="city", y="temperature")
chart.save("temperature.pdf")
table.save("weather.xlsx")
```

> Question, test, figure, saved table: the whole analysis in eight lines.

Back to day 13

Day 14: Review of the second week

1. Which word lists every change you made to a table, for your methods? (day 13)
   → `table.steps()`
2. Which word makes all the text on a chart bigger or smaller? (day 1)
   → `chart.font_size(14)`
3. A word for one axis ends in ...? (day 2)
   → _x or _y: `label_y`, `range_x`, `log_scale_y`.
4. Which word turns a column of text such as `"144"` into numbers? (day 8)
   → `table.make_numbers("height")`
5. Which word sets the y-axis from 0 to 100? (day 4)
   → `chart.range_y(0, 100)`
6. Which word shows the rows with outliers? (day 11)
   → `table.show_rows_with_outliers("marks", for_each="class")`
7. Which word joins two tables on a column they share? (day 9)
   → `table.join_with(ages, using="name")`
8. How do you save a picture sharp enough for a journal? (day 5)
   → `chart.save("figure.tif", resolution=600)`, or save it as `.pdf`

Task 1. School table: count the empty cells, remove the rows with an empty `"sleep"`, give the average sleep for each value of `"passed"`, test the difference, and draw boxes with all the dots.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.count_empty_cells()
table.remove_rows_with_empty_cells("sleep")
table.average_of("sleep", for_each="passed")
table.compare_averages("sleep", for_each="passed")
chart = table.chart()
chart.boxes(x="passed", y="sleep")
chart.all_dots(x="passed", y="sleep")
chart.save("sleep.png")
```

> Clean, sum up, test, draw: the order of every analysis.

Back to day 14

Day 15: Models: explain one column from several

1. How do you give the panels of a figure the letters A and B? (day 13)
   → `letters=True` in `sdata.grid(...)` or `sdata.side_by_side(...)`
2. Which word compares one average with a number you choose? (day 12)
   → `table.compare_average_to("marks", 60)`
3. Which word adds a new column made from other columns? (day 8)
   → `table.add_column("total", "Maths + Science")`, or `table.add_column("per_hour", by_dividing=["marks", "hours"])`
4. How do you start a table from the marks practice table? (day 1)
   → table = `sdata.table(sdata.example("marks"))`

Task 1. School table: explain `"days_absent"` using `"sleep"` and `"breakfast"` with `outcome="counts"`. Does breakfast matter?

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.explain("days_absent", using=["sleep", "breakfast"], outcome="counts")
```

> Counts of days call for Poisson regression; read the line for breakfast and its p-value.

Back to day 15

Day 16: Patterns across many columns

1. Which word explains marks from hours, sleep and breakfast together? (day 15)
   → `table.explain("marks", using=["hours", "sleep", "breakfast"])`
2. Which word hands a table on to pandas? (day 13)
   → `table.as_pandas()`
3. Which word gives several summaries at once? (day 9)
   → `table.summary_of("sales", for_each="shop", show=["average", "median", "count"])`
4. Which file endings give a picture that stays sharp at any size? (day 2)
   → `.pdf` and `.svg`

Task 1. Meadow table: add 3 clusters and count the rows of each cluster across `"field"`.

```
import simplydata as sdata
table = sdata.table(sdata.example("meadow"))
table.add_clusters(how_many=3)
table.count_rows(for_each="cluster", across="field")
```

> If each cluster is mostly one field, the fields have different plants.

Back to day 16

Day 17: Survival, and planning a study

1. Which word adds the principal components PC1 and PC2? (day 16)
   → `table.add_principal_components(how_many=2)`
2. Which word lists every change you made to a table, for your methods? (day 13)
   → `table.steps()`
3. A p-value answers `"could it be chance?"`. What answers `"how big is it?"` (day 10)
   → The effect size: the difference in averages, Cohen's d or Hedges' g, omega-squared.
4. Which two words show the spread of each group, with every value as a dot on top? (day 3)
   → `chart.boxes(x="class", y="marks")` and `chart.all_dots(x="class", y="marks")`

Task 1. How many values does each group need to find a large difference (`effect_size=0.8`) with power 0.8? And with a small one (0.2)?

```
import simplydata as sdata
sdata.how_many_needed(effect_size=0.8, power=0.8)
sdata.how_many_needed(effect_size=0.2, power=0.8)
```

About 26 for a large difference but about 394 for a small one: small effects need big studies.

Back to day 17

Day 18: Fitting lines, curves and formulas

1. Which word draws survival curves? (day 17)
   → `chart.survival_curves(time="days", event="died", colour="treatment")`
2. How do you explain a yes-or-no column such as `"passed"`? (day 15)
   → `outcome="yes or no"` (logistic regression)
3. Which word checks whether values are bell-shaped (normal)? (day 11)
   → `table.compare_shape_to("marks", "normal")`
4. Which word draws a line across at y = 50, with a label? (day 4)
   → `chart.line_at_y(50, label="pass mark")`

Task 1. Growth table: fit the formula `"forest = a + b * (year - 2000)"` for each country. Which country loses forest fastest?

```
import simplydata as sdata
table = sdata.table(sdata.example("growth"))
table.fit_formula("forest = a + b * (year - 2000)", for_each="country")
```

b is the change in forest each year: the country with the most negative b loses forest fastest.

Back to day 18

Day 19: Finding, solving and distributions

1. Which word fits a straight line on a table? (day 18)
   → `table.fit_line(x="hours", y="marks")`
2. Which word puts rows that are alike into groups? (day 16)
   → `table.add_clusters(how_many=2)`
3. What should a results sentence give? (day 12)
   → The summary of each group with n, the effect size (with its interval), the name of the test and the exact p-value.
4. Which word writes a note with an arrow on the chart? (day 5)
   → `chart.note("X marks the spot", x=10, y=4, arrow=True)`

Task 1. A box has a square bottom of side x and a height of 10 - x, so its volume is x * x * (10 - x). For which x between 0 and 10 is the volume biggest?

```
import simplydata as sdata
sdata.find_biggest("x * x * (10 - x)", between=[0, 10])
```

x = 6.67 (that is, 20/3), with a volume of about 148.

Task 2. Heights are normal with average 150 and standard deviation 8. What share is taller than 165? Which height do 95 % of them stay below?

```
import simplydata as sdata
heights = sdata.distribution("normal", average=150, standard_deviation=8)
heights.chance(above=165)
heights.percentile(95)
```

About 3 % are taller than 165, and 95 % are below about 163.

Back to day 19

Day 20: Values between points

1. Which word finds where a formula is smallest? (day 19)
   → `sdata.find_smallest("x**2 - 4*x + 1", between=[0, 5])`
2. Which word tests whether survival differs between groups? (day 17)
   → `table.compare_survival(time="days", event="died", for_each="treatment")`
3. How do you give the panels of a figure the letters A and B? (day 13)
   → `letters=True` in `sdata.grid(...)` or `sdata.side_by_side(...)`
4. Which word sorts with the biggest marks first? (day 6)
   → `table.sort_by("marks", biggest_first=True)`

Task 1. Weather table: estimate the `"rain"` of each city at `month_number` 2.5 and 7.5, joined by straight lines.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
rain = table.values_at("rain", at={"month_number": [2.5, 7.5]}, joined_by="straight lines", for_each="city")
rain.show()
```

> Each estimate lies halfway between the two months around it.

Back to day 20

Day 21: Review, and the question of your project

1. Which word estimates a column between the points you measured? (day 20)
   → `table.values_at("population", at={"year": [2001, 2011]})`
2. Which word cuts a column of numbers into named groups? (day 8)
   → `table.add_group_column("grade", from_column="marks", cut_at=[40, 60, 80], names=["D", "C", "B", "A"])`
3. Which word keeps only the rows with marks above 50? (day 1)
   → `table.keep_rows("marks", above=50)`
4. How do you make simplydata numbers from a list? (day 9)
   → nums = `sdata.numbers([12, 15, 9, 20, 14])`
5. Which option gives the rank-based test, for small groups that are not bell-shaped? (day 10)
   → `based_on="ranks"`
6. Which word adds the values a model predicts as a new column? (day 15)
   → `table.add_predicted_column("marks", using=["hours", "breakfast"])`
7. You did many tests at once. Which word corrects the p-values? (day 11)
   → `add_adjusted_p_value_column(method="holm")`, on the table of results
8. A few values are much bigger than the rest. Which word helps? (day 4)
   → `chart.log_scale_y()`
9. How do you fit a curve with x and x squared? (day 18)
   → `table.fit_curve(x="hours", y="marks", degree=2)`

Task 1. Trial table: test whether survival differs between the treatments, then explain the risk of dying from `"treatment"` and `"age"` together.

```
import simplydata as sdata
table = sdata.table(sdata.example("trial"))
table.compare_survival(time="days", event="died", for_each="treatment")
table.explain_survival(time="days", event="died", using=["treatment", "age"])
```

> The log-rank test compares the curves; the Cox model gives a hazard ratio for each column.

Task 2. Write your question in one sentence. Open your raw file and write down how many rows and columns it has, and which columns have empty cells.

```
import simplydata as sdata
sdata.table(sdata.example("school")).save("school_raw.csv")
table = sdata.table("school_raw.csv")
table.count_rows()
table.describe_columns()
```

> The first line only makes a raw file to practise on; with your own data, start at the second line. The school table has 180 rows and 8 columns; sleep has 19 empty cells.

Back to day 21

Day 22: Project: clean

1. Which word tests whether groups differ in their make-up across many columns (PERMANOVA)? (day 16)
   → `table.compare_profiles(for_each="grazed")`
2. Which word finds where the two sides of an equation are equal? (day 19)
   → `sdata.solve("x**3 - 2*x = 5", between=[0, 3])`
3. Which word says how many values you need in each group? (day 17)
   → `sdata.how_many_needed(effect_size=0.5, power=0.8)`
4. You made a mistake in the last step. Which word takes it back? (day 6)
   → `table.undo()`; `table.start_again()` goes back to the start.

Task 1. Clean your table and save it as a new file. Write one line for each decision you made about empty cells.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.count_empty_cells()
table.add_is_empty_column("sleep_was_empty", from_column="sleep")
table.rename_column("days_absent", to="absent")
table.save("school_clean.csv")
table.steps()
```

> Here the empty sleep cells are kept (tests skip them, and models can use multiple imputation); a column remembers which were empty.

Back to day 22

Day 23: Project: explore

1. Which word fills an even map from scattered points, ready for a contour map? (day 20)
   → `table.map_of("height", across=["east", "north"])`
2. Which word fits a formula of your own, such as growth? (day 18)
   → `table.fit_formula("population = a * exp(b * (year - 2000))")`
3. Which word gives the middle value of numbers? (day 9)
   → `nums.median()`
4. Which word shows the numbers on the y-axis as percents? (day 4)
   → `chart.tick_format_y("percent")`

Task 1. Make three summaries and two charts of your table that help to answer your question.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.count_each("breakfast")
table.summary_of("marks", for_each="breakfast", show=["average", "median", "standard deviation", "count"])
table.correlation_between("sleep", "marks")
chart = table.chart()
chart.histogram(x="marks", colour="breakfast")
chart.save("marks_histogram.png")
sdata.pairs(table, colour="breakfast").save("pairs.png")
```

> Each summary and chart is a small answer; the test comes later.

Back to day 23

Day 24: Project: figures

1. Which word gives the average mark of each class? (day 1)
   → `table.average_of("marks", for_each="class")`
2. How do you make a normal distribution with average 50 and standard deviation 10? (day 19)
   → `sdata.distribution("normal", average=50, standard_deviation=10)`
3. Which word keeps rows 10 to 20, by their place in the table? (day 6)
   → `table.keep_rows_by_position(10, to=20)`
4. How do you make 100 random numbers with average 50 and standard deviation 10, the same every time? (day 9)
   → `sdata.random_numbers(how_many=100, average=50, standard_deviation=10, seed=1)`

Task 1. Make your main figure with two panels and save it ready for a journal.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
a = table.chart()
a.boxes(x="breakfast", y="marks")
a.all_dots(x="breakfast", y="marks")
a.label_y("Mark (out of 100)")
b = table.chart()
b.dots(x="sleep", y="marks", colour="breakfast")
b.fit_line(confidence_band=True)
b.label_x("Sleep (hours)")
figure = sdata.grid(a, b, per_row=2, letters=True)
figure.style("paper")
figure.size(width=17, height=8, unit="cm")
figure.save("figure1.pdf")
figure.save("figure1.tif", resolution=600)
```

> Panel A shows the groups, panel B the link with sleep; the band is the 95 % confidence interval.

Back to day 24

Day 25: Project: tests

1. Which word gives the chance of a value above 80? (day 19)
   → `dist.chance(above=80)`
2. How do you get the number in row 1, column 2 of a grid? (day 9)
   → `grid.number_at(row=1, column=2)`
3. Which word lists every step you did to a table? (day 6)
   → `table.steps()`
4. How do you give each class its own colour in a bar chart? (day 2)
   → `colour="class"` inside the chart step: `chart.bars(x="subject", y="marks", colour="class")`

Task 1. Run the test that answers your question. Write down the test's name, the effect size and the p-value.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.compare_shape_to("marks", "normal", for_each="breakfast")
table.compare_averages("marks", for_each="breakfast")
table.correlation_between("sleep", "marks")
```

> Two groups of numbers: Welch's t-test. Two columns of numbers: Pearson's correlation.

Back to day 25

Day 26: Project: models and fits

1. Which word fits a distribution to a column? (day 19)
   → `table.fit_distribution("days_absent", "negative binomial")`
2. Which word draws one dot for each row, x against y? (day 3)
   → `chart.dots(x="hours", y="marks")`
3. The same children were measured twice. Which word compares the two? (day 12)
   → `table.compare_before_after("before", "after")`: the paired test.

4. Which word draws one small chart for each class? (day 5)
   → `chart.one_chart_per("class")`

Task 1. Explain your main column from two or three others, and say in one sentence what changed compared with the single test.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.explain("marks", using=["hours", "sleep", "breakfast"])
table.fit_line(x="sleep", y="marks")
```

> Each line of the model is the effect of one column with the others held the same.

Back to day 26

Day 27: Project: results text

1. Which word hands a table on to pandas? (day 13)
   → `table.as_pandas()`
2. Which word explains marks from hours, sleep and breakfast together? (day 15)
   → `table.explain("marks", using=["hours", "sleep", "breakfast"])`
3. Which word tests whether the counts of categories differ between groups? (day 11)
   → `table.compare_counts("result", for_each="class")`
4. Which word tests whether the average marks differ between the classes? (day 10)
   → `table.compare_averages("marks", for_each="class")`

Task 1. Write a results paragraph, a methods paragraph and a figure caption for your project.

> For example: `"Pupils who had breakfast scored on average 3.4 marks more than pupils who did not (49.7 against 46.3; n = 115 and 65), but a difference this big could be chance (Welch's t-test, p = 0.051; Hedges' g = 0.31). Sleep was not recorded for 19 pupils; they were left out of the tests that use sleep. "` Use your own numbers, exactly as printed, and report a result that could be chance as plainly as one that could not.

Back to day 27

Day 28: Review, and finish your project

1. Which word counts the empty cells in each column? (day 8)
   → `table.count_empty_cells()`
2. Which word draws survival curves? (day 17)
   → `chart.survival_curves(time="days", event="died", colour="treatment")`
3. Which word adds the principal components PC1 and PC2? (day 16)
   → `table.add_principal_components(how_many=2)`
4. Which word estimates a column between the points you measured? (day 20)
   → `table.values_at("population", at={"year": [2001, 2011]})`
5. Which word fits a straight line on a table? (day 18)
   → `table.fit_line(x="hours", y="marks")`
6. Which option makes a bar show the median instead of the average? (day 4)
   → `show="median"`
7. Which word lists the practice tables? (day 1)
   → `sdata.example()`; `sdata.example("marks")` gets one of them.
8. Which word gives the average of a column for each group? (day 9)
   → `table.average_of("marks", for_each="class")`
9. How do you look at the first 5 rows of a table? (day 6)
   → `table.show_first_rows(5)`
10. Which word moves the legend below the chart? (day 2)
    → `chart.legend_bottom()`

Task 1. Without looking back: which word fits a formula, which finds where it is biggest, and which gives the chance of a value above a number? Use one of them on any table or formula.

```
import simplydata as sdata
```

```
dist = sdata.distribution("normal", average=50, standard_deviation=10)
dist.chance(above=70)
sdata.find_biggest("x * (10 - x)", between=[0, 10])
```

`fit_formula`, `find_biggest` and chance: three families (`fit_`, `find_`, and the distribution words).

Task 2. Run your whole project script from the top. Check that it makes the clean table, the figure and the numbers in your text.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.add_is_empty_column("sleep_was_empty", from_column="sleep")
table.compare_averages("marks", for_each="breakfast")
chart = table.chart()
chart.boxes(x="breakfast", y="marks")
chart.save("figure1.pdf")
table.save("school_clean.csv")
table.steps()
```

One script from the raw file to the figure: anyone can run it again and get the same answers.

Back to day 28

Day 29: Final test from memory

1. Which word finds where a formula is smallest? (day 19)
   → `sdata.find_smallest("x**2 - 4*x + 1", between=[0, 5])`
2. Which word compares one average with a number you choose? (day 12)
   → `table.compare_average_to("marks", 60)`
3. Which word puts two charts next to each other on one page? (day 5)
   → `sdata.side_by_side(a, b)`
4. Which word draws a line with a dot at each point? (day 3)
   → `chart.lines(x="month", y="temperature", dots=True)`
5. How do you explain a yes-or-no column such as `"passed"`? (day 15)
   → `outcome="yes or no"` (logistic regression)
6. Which word lists every change you made to a table, for your methods? (day 13)
   → `table.steps()`
7. Which word shows the rows with outliers? (day 11)
   → `table.show_rows_with_outliers("marks", for_each="class")`
8. Which word says which pairs of groups differ? (day 10)
   → `table.compare_each_pair("marks", for_each="class")`
9. Which word fills an even map from scattered points, ready for a contour map? (day 20)
   → `table.map_of("height", across=["east", "north"])`
10. How do you fit a curve with x and x squared? (day 18)
    → `table.fit_curve(x="hours", y="marks", degree=2)`
11. Which word tests whether survival differs between groups? (day 17)
    → `table.compare_survival(time="days", event="died", for_each="treatment")`
12. Which word puts rows that are alike into groups? (day 16)
    → `table.add_clusters(how_many=2)`
13. Which word tells whether two columns go up and down together? (day 9)
    → `table.correlation_between("hours", "marks")`
14. How do you fill the empty marks with the median? (day 8)
    → `table.fill_empty_cells("marks", with_the="median")`

Task 1. From memory, in an empty file: open the `"marks"` table, remove the rows with empty cells, give the average marks for each class, test the difference, and save a two-panel figure (boxes, and dots of hours against marks with a fitted line) as a PDF.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.remove_rows_with_empty_cells()
table.average_of("marks", for_each="class")
```

```
table.compare_averages("marks", for_each="class")
a = table.chart()
a.boxes(x="class", y="marks")
b = table.chart()
b.dots(x="hours", y="marks")
b.fit_line()
figure = sdata.grid(a, b, per_row=2, letters=True)
figure.save("final.pdf")
```

If you wrote this from memory, you know the core of simplydata. Words you missed: run their `sdata.help` again tomorrow.

Back to day 29

Day 30: The same work in pandas and matplotlib

1. Which word estimates a column between the points you measured? (day 20)
   → `table.values_at("population", at={"year": [2001, 2011]})`
2. Which word finds where the two sides of an equation are equal? (day 19)
   → `sdata.solve("x**3 - 2*x = 5", between=[0, 3])`
3. Which word fits a formula of your own, such as growth? (day 18)
   → `table.fit_formula("population = a * exp(b * (year - 2000))")`
4. Which word tests whether groups differ in their make-up across many columns (PERMANOVA)? (day 16)
   → `table.compare_profiles(for_each="grazed")`

Task 1. In simplydata: keep classes A and B of the marks table, give the average marks of each class, test the difference and draw it as bars.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("class", one_of=["A", "B"])
table.average_of("marks", for_each="class")
table.compare_averages("marks", for_each="class")
chart = table.chart()
chart.bars(x="class", y="marks")
chart.save("simplydata_bars.png")
```

The simplydata version: one word per step, and each answer printed in a sentence.

Task 2. Write the same analysis with pandas, SciPy and matplotlib, then compare the two line by line.

```
import matplotlib.pyplot as plt
import simplydata as sdata
from scipy import stats
df = sdata.example("marks")
df = df[df["class"].isin(["A", "B"])]
print(df.groupby("class")["marks"].mean())
a = df.loc[df["class"] == "A", "marks"]
b = df.loc[df["class"] == "B", "marks"]
print(stats.ttest_ind(a, b, equal_var=False))
fig, ax = plt.subplots()
means = df.groupby("class")["marks"].mean()
ax.bar(means.index, means.values)
fig.savefig("pandas_bars.png")
```

The same in pandas, SciPy and matplotlib: `equal_var=False` makes it Welch's t-test, as in simplydata.

Back to day 30

Answers: Beginner track

The warm-up answers and a worked answer to each task

There is often more than one good answer: if yours runs and answers the question, it is right. Every piece of code here runs as it is (the tests run it).

Day 1: Python first steps: values and names

Task 1. Give the text `"Bergen"` the name city and the number 2250 the name rain (the rain in a year, in mm). Print the words `"Rain in"`, the city, and the rain of an average month (rain / 12).

```
city = "Bergen"
rain = 2250
print("Rain in", city, rain / 12)
```

> It prints: Rain in Bergen 187.5. Text in quotes, numbers without, and print with commas between the values.

Back to day 1

Day 2: Python first steps: lists and words

1. How do you write the text Asha in Python? (day 1)
   → `"Asha"`, in quotes. Without quotes, Python thinks Asha is a name.
2. How do you give the number 12 the name age? (day 1)
   → age = 12. The name goes on the left of =, the value on the right.
3. What does `print(age)` do? (day 1)
   → It shows the value of age on the screen.

Task 1. Make a list of five temperatures, turn it into simplydata numbers, and find the biggest one and the median.

```
import simplydata as sdata
temperatures = [4, 9, 13, 17, 20]
nums = sdata.numbers(temperatures)
nums.biggest()
nums.median()
```

> The biggest is 20 and the median (the middle value when they are in order) is 13.

Back to day 2

Day 3: Reading an error, and your first chart

1. How do you write a list of the numbers 3, 5 and 2? (day 2)
   → [3, 5, 2]: square brackets, commas between.
2. In `nums.keep_numbers(above=50)`, what is `above=50`? (day 2)
   → An option with a name: the name, =, and the value, inside the round brackets.
3. How do you write the text Asha in Python? (day 1)
   → `"Asha"`, in quotes. Without quotes, Python thinks Asha is a name.

Task 1. Find and fix the two mistakes in these lines, then run them: nums = `sdata.numbers([4, 8, 6)` and then `nums.averag()`

```
import simplydata as sdata
nums = sdata.numbers([4, 8, 6])
nums.average()
```

> The list needs its closing] before the), and the word is average. The average is 6.

Task 2. Make a bar chart of the `"weather"` table with `x="city"` and `y="rain"`, and save it as `"rain.png"`.

```
import simplydata as sdata
weather = sdata.example("weather")
```

```
chart = sdata.chart(weather)
chart.bars(x="city", y="rain")
chart.save("rain.png")
```

One bar for each city, as tall as its average rain over the 12 months.

Back to day 3

Day 4: Titles and labels

1. Where in an error message do you look first? (day 3)
   → At the last line: it names the kind of error and says what went wrong.
2. How do you give the number 12 the name age? (day 1)
   → age = 12. The name goes on the left of =, the value on the right.
3. What does NameError: name 'nmae' is not defined usually mean? (day 3)
   → A name is misspelt, or was never given a value (with =) before it was used.

Task 1. Draw dots of `"hours"` (x) against `"marks"` (y) from the marks table, with a title, a name for each axis and a caption at the bottom.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.title_top("More study, better marks?")
chart.label_x("Hours of study")
chart.label_y("Mark (out of 100)")
chart.caption_bottom("Made-up data")
chart.save("hours_and_marks.png")
```

First draw, then add the words; the order of the word steps does not matter.

Back to day 4

Day 5: Colours, groups and legends

1. Which word puts a title above the chart? (day 4)
   → `chart.title_top("My title")`
2. What does every simplydata program start with? (day 2)
   → import simplydata as sdata
3. How do you name the y-axis? (day 4)
   → `chart.label_y("Marks out of 100")`

Task 1. Weather table: draw lines of `"temperature"` over `"month"`, one colour for each city. Put the legend on top and use the style `"paper"`.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city")
chart.legend_top()
chart.style("paper")
chart.save("temperature.png")
```

`colour="city"` draws one line for each city and adds a legend; the style changes the whole look.

Back to day 5

Day 6: Different chart types

1. How do you give each class its own colour in a bar chart? (day 5)
   → `colour="class"` inside the chart step: `chart.bars(x="subject", y="marks", colour="class")`
2. Python says SyntaxError. What do you check first? (day 3)
   → Quote marks and brackets: every one that opens must be closed.

3. Which word moves the legend below the chart? (day 5)
   → `chart.legend_bottom()`

Task 1. Draw violins of `"temperature"` for each city in the weather table, with all the dots on top.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.violins(x="city", y="temperature")
chart.all_dots(x="city", y="temperature")
chart.save("violins.png")
```

> A violin shows the shape of the spread; the dots show every month.

Back to day 6

Day 7: Review of the first week

1. Which chart shows how the values of one column are spread, in bins? (day 6)
   → `chart.histogram(x="marks", bins=8)`
2. The names under the bars bump into each other. Which word helps? (day 4)
   → `chart.rotate_labels_x(45)`
3. Which word gives the average of numbers called nums? (day 2)
   → `nums.average()`
4. Which word lets you choose the colours of the groups? (day 5)
   → `chart.colours("orange", "teal", "purple")`
5. Which word lists the practice tables? (day 3)
   → `sdata.example()`; `sdata.example("marks")` gets one of them.
6. What does `print(age)` do? (day 1)
   → It shows the value of age on the screen.

Task 1. Make numbers of five marks and give their average. Then draw bars of the marks table with `x="subject"`, `y="marks"`, one colour for each class, and a title. Save it.

```
import simplydata as sdata
nums = sdata.numbers([48, 71, 55, 88, 62])
nums.average()
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks", colour="class")
chart.title_top("Marks in each subject")
chart.save("subjects.png")
```

> Python lists (day 2), bars and save (day 3), titles (day 4) and colours (day 5).

Back to day 7

Day 8: Averages, error bars and fit lines

1. Which two words show the spread of each group, with every value as a dot on top? (day 6)
   → `chart.boxes(x="class", y="marks")` and `chart.all_dots(x="class", y="marks")`
2. How do you put the bars in your own order? (day 5)
   → `chart.order_x("C", "B", "A")`
3. How do you write the text Asha in Python? (day 1)
   → `"Asha"`, in quotes. Without quotes, Python thinks Asha is a name.

Task 1. Weather table: draw bars of the average `"rain"` for each city with 95 % confidence intervals, and write the numbers on the bars.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.bars(x="city", y="rain", error="confidence interval")
chart.numbers_on_bars()
```

```
chart.save("rain_ci.png")
```

A line is printed saying which error bars were drawn, so the caption can say it too.

Back to day 8

Day 9: Axes and reference lines

1. Which option makes a bar show the median instead of the average? (day 8)
   → `show="median"`
2. Which word draws one dot for each row, x against y? (day 6)
   → `chart.dots(x="hours", y="marks")`
3. How do you write a list of the numbers 3, 5 and 2? (day 2)
   → [3, 5, 2]: square brackets, commas between.

Task 1. Growth table: draw lines of `"population"` over `"year"`, one for each country, with a line at x = 2010 and the y-axis starting at 0.

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.lines(x="year", y="population", colour="country")
chart.line_at_x(2010, label="2010")
chart.range_y(0, 60)
chart.save("growth.png")
```

`range_y` takes the lowest and the highest value of the axis; `line_at_x` draws a line going up.

Back to day 9

Day 10: Several charts together

1. Which word sets the y-axis from 0 to 100? (day 9)
   → `chart.range_y(0, 100)`
2. Which word draws a line with a dot at each point? (day 6)
   → `chart.lines(x="month", y="temperature", dots=True)`
3. How do you start a chart of a table called marks? (day 3)
   → chart = `sdata.chart(marks)`

Task 1. Make a histogram and a box chart of the marks, and put them one above the other on one page.

```
import simplydata as sdata
marks = sdata.example("marks")
a = sdata.chart(marks).histogram(x="marks")
b = sdata.chart(marks).boxes(x="class", y="marks")
page = sdata.one_above_other(a, b)
page.save("two_charts.png")
```

Each chart is made first; `one_above_other` puts them on one page.

Back to day 10

Day 11: A figure for a journal, and word families

1. Which word draws one small chart for each class? (day 10)
   → `chart.one_chart_per("class")`
2. Which option adds error bars that say how sure we are of the average? (day 8)
   → `error="standard error"`, or `error="confidence interval"` for the 95 % interval.
3. Which word makes all the text on a chart bigger or smaller? (day 4)
   → `chart.font_size(14)`

Task 1. A figure for a journal: dots of `"hours"` against `"marks"`, style `"paper"`, 8.5 cm wide and 7 cm high, font size 8, saved as a `.tif` with 600 dpi.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.style("paper")
chart.size(width=8.5, height=7, unit="cm")
chart.font_size(8)
chart.save("figure1.tif", resolution=600)
```

> Journals often ask for 8.5 cm (one column) or 17 cm (two columns) and 300 to 600 dpi.

Task 2. Save one chart as `.png`, `.pdf` and `.svg`. Then guess, before you look anything up: which word hides the legend, and which puts it on the left? Try your guesses on a chart with `colour=`.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks", colour="class")
chart.save("subjects.png")
chart.save("subjects.pdf")
chart.save("subjects.svg")
chart.legend_left()
chart.hide_legend()
chart.save("no_legend.png")
```

> `legend_left` follows `legend_bottom` and `legend_top`; `hide_legend` follows `show_` and `hide_`. A family is learned once.

Back to day 11

Day 12: Open a table and look at it

1. Which word adds a second y-axis on the right? (day 11)
   → `chart.second_y("rain")`
2. Which word draws a line across at y = 50, with a label? (day 9)
   → `chart.line_at_y(50, label="pass mark")`
3. How do you give each class its own colour in a bar chart? (day 5)
   → `colour="class"` inside the chart step: `chart.bars(x="subject", y="marks", colour="class")`

Task 1. Open the `"shop"` practice table. How many rows does it have, and how many rows does each shop have?

```
import simplydata as sdata
table = sdata.table(sdata.example("shop"))
table.count_rows()
table.count_each("shop")
table.describe_columns()
```

> `count_rows` counts all the rows, `count_each` the rows of each shop; `describe_columns` also shows the empty cells.

Back to day 12

Day 13: Choose rows and columns, and undo

1. How do you look at the first 5 rows of a table? (day 12)
   → `table.show_first_rows(5)`
2. Which word puts two charts next to each other on one page? (day 10)
   → `sdata.side_by_side(a, b)`
3. Which chart shows how the values of one column are spread, in bins? (day 6)
   → `chart.histogram(x="marks", bins=8)`

Task 1. Weather table: keep the city `"Sunnyvale"`, keep only the columns month, temperature and rain, sort by rain with the biggest first, and save it as `"sunny.csv"`.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.keep_rows("city", equal_to="Sunnyvale")
table.keep_columns("month", "temperature", "rain")
table.sort_by("rain", biggest_first=True)
table.save("sunny.csv")
table.steps()
```

Each step prints what it did; `steps()` lists them all, in order.

Back to day 13

Day 14: Review of the second week

1. Which word keeps only the rows where subject is Maths? (day 13)
   → `table.keep_rows("subject", equal_to="Maths")`
2. Which word draws one bar for each name, as tall as the marks? (day 3)
   → `chart.bars(x="name", y="marks")`
3. Which word draws a straight line through the dots? (day 8)
   → `chart.fit_line()`, after `chart.dots(...)`
4. Which word sets the size of a figure in cm? (day 11)
   → `chart.size(width=17, height=8, unit="cm")`
5. A few values are much bigger than the rest. Which word helps? (day 9)
   → `chart.log_scale_y()`
6. Which word describes each column: its kind, empty cells and numbers? (day 12)
   → `table.describe_columns()`
7. How do you put four charts in two rows, with the letters A to D? (day 10)
   → `sdata.grid(a, b, c, d, per_row=2, letters=True)`
8. How do you give the number 12 the name age? (day 1)
   → age = 12. The name goes on the left of =, the value on the right.

Task 1. Weather table: keep the city `"Frostby"`, sort by `"temperature"`, and draw a line of the temperature over the month, with a line at y = 0 and the label `"freezing"`.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.keep_rows("city", equal_to="Frostby")
table.sort_by("temperature")
table.show_first_rows(3)
chart = table.chart()
chart.lines(x="month_number", y="temperature")
chart.line_at_y(0, label="freezing")
chart.save("frostby.png")
```

Tables (days 12 and 13) feed charts (days 3 to 11).

Back to day 14

Day 15: Clean the data and add columns

1. Which word sorts with the biggest marks first? (day 13)
   → `table.sort_by("marks", biggest_first=True)`
2. Which word counts the rows of each fruit? (day 12)
   → `table.count_each("fruit")`
3. Which word checks by eye whether values are bell-shaped? (day 8)
   → `chart.qq_dots("marks")`
4. What does `print(age)` do? (day 1)
   → It shows the value of age on the screen.

Task 1. Shop table: count the empty cells, fill the empty `"sales"` with the line between neighbours for each shop (sorted by `"date"`), and add a column `"weekday"` with the name of the weekday.

```
import simplydata as sdata
table = sdata.table(sdata.example("shop"))
table.count_empty_cells()
table.fill_empty_cells("sales", with_the="line between neighbours", sorted_by="date", for_each="shop")
table.add_date_part_column("weekday", from_column="date", part="weekday name")
table.show_first_rows()
```

`for_each=` keeps the two shops apart, so a gap is never filled from the other shop's days.

Back to day 15

Day 16: Summaries, and from a table to a chart

1. Which word counts the empty cells in each column? (day 15)
   → `table.count_empty_cells()`
2. You made a mistake in the last step. Which word takes it back? (day 13)
   → `table.undo()`; `table.start_again()` goes back to the start.
3. Which word shows the numbers on the y-axis as percents? (day 9)
   → `chart.tick_format_y("percent")`
4. In `nums.keep_numbers(above=50)`, what is `above=50`? (day 2)
   → An option with a name: the name, =, and the value, inside the round brackets.

Task 1. Fruit table: the total `"count"` of each fruit, drawn as bars with the numbers on the bars.

```
import simplydata as sdata
table = sdata.table(sdata.example("fruit"))
totals = table.total_of("count", for_each="fruit")
chart = totals.chart()
chart.bars(x="fruit", y="total of count")
chart.numbers_on_bars()
chart.save("fruit_totals.png")
```

A summary gives back a new table, with a column called `"total of count"`, ready for a chart.

Back to day 16

Day 17: Lists and grids of numbers

1. Which word gives the average of a column for each group? (day 16)
   → `table.average_of("marks", for_each="class")`
2. Which word keeps rows 10 to 20, by their place in the table? (day 13)
   → `table.keep_rows_by_position(10, to=20)`
3. Which word draws one small chart for each class? (day 10)
   → `chart.one_chart_per("class")`
4. How do you save a chart as a PDF? (day 3)
   → `chart.save("my_chart.pdf")`: the end of the file name decides the type.

Task 1. Make 200 random heights with average 150 and standard deviation 8 (`seed=1`). What is their 90th percentile? Draw them as a histogram.

```
import simplydata as sdata
heights = sdata.random_numbers(how_many=200, average=150, standard_deviation=8, seed=1)
heights.percentile(90)
chart = heights.chart()
chart.histogram(x="value")
chart.save("heights.png")
```

About 157: nine heights in ten are below it. `seed=1` gives the same numbers every time.

Back to day 17

Day 18: Is a difference real? Tests and effect sizes

1. How do you make simplydata numbers from a list? (day 17)
   → nums = `sdata.numbers([12, 15, 9, 20, 14])`
2. How do you fill the empty marks with the median? (day 15)
   → `table.fill_empty_cells("marks", with_the="median")`
3. How do you save a picture sharp enough for a journal? (day 11)
   → `chart.save("figure.tif", resolution=600)`, or save it as `.pdf`
4. Which word puts a title above the chart? (day 4)
   → `chart.title_top("My title")`

Task 1. School table: do the average marks differ between pupils who had breakfast and pupils who did not? Write down the difference, the effect size and the p-value.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.average_of("marks", for_each="breakfast")
table.compare_averages("marks", for_each="breakfast")
```

> Two groups, so it is Welch's t-test; the sentence gives the difference, Hedges' g and the p-value.

Back to day 18

Day 19: Counts, outliers and many tests

1. Which word tests whether the average marks differ between the classes? (day 18)
   → `table.compare_averages("marks", for_each="class")`
2. Which word tells whether two columns go up and down together? (day 16)
   → `table.correlation_between("hours", "marks")`
3. Which word counts the rows? (day 12)
   → `table.count_rows()`
4. Which word moves the legend below the chart? (day 5)
   → `chart.legend_bottom()`

Task 1. School table: is the share of pupils who `"passed"` different with and without breakfast?

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.count_rows(for_each="breakfast", across="passed")
table.compare_counts("passed", for_each="breakfast")
```

> passed is a category (yes or no), so the test compares counts: the chi-square test, with Cramer's V as the effect size.

Back to day 19

Day 20: Choosing a test, and a whole analysis

1. Which word tests whether the counts of categories differ between groups? (day 19)
   → `table.compare_counts("result", for_each="class")`
2. Which word gives the middle value of numbers? (day 17)
   → `nums.median()`
3. Which word lists every step you did to a table? (day 13)
   → `table.steps()`
4. Which two words show the spread of each group, with every value as a dot on top? (day 6)
   → `chart.boxes(x="class", y="marks")` and `chart.all_dots(x="class", y="marks")`

Task 1. Marks table: is each child's Maths mark different from the same child's Science mark? Choose the right word first.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_columns("name", "subject", "marks")
table.turn_rows_into_columns(names_from="subject", values_from="marks")
```

```
table.compare_before_after("Maths", "Science")
```

> The same children twice, so the test must be paired. First one row per child, with a Maths and a Science column.

Task 2. School table: is the average `"sleep"` different from 8 hours?

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.compare_average_to("sleep", 8)
```

> One average against a number you choose: the one-sample t-test. Empty sleep cells are skipped, with a note.

Task 3. From memory: weather table, compare the average `"temperature"` of the cities, draw boxes with all the dots, and save the figure as a PDF and the table as `"weather.xlsx"`.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.compare_averages("temperature", for_each="city")
chart = table.chart()
chart.boxes(x="city", y="temperature")
chart.all_dots(x="city", y="temperature")
chart.save("temperature.pdf")
table.save("weather.xlsx")
```

> Question, test, figure, saved table: the whole analysis in eight lines.

Back to day 20

Day 21: Review of the third week

1. The same children were measured twice. Which word compares the two? (day 20)
   → `table.compare_before_after("before", "after")`: the paired test.
2. Which word turns a column of text such as `"144"` into numbers? (day 15)
   → `table.make_numbers("height")`
3. Which word writes a note with an arrow on the chart? (day 11)
   → `chart.note("X marks the spot", x=10, y=4, arrow=True)`
4. Which word says which pairs of groups differ? (day 18)
   → `table.compare_each_pair("marks", for_each="class")`
5. Which word joins two tables on a column they share? (day 16)
   → `table.join_with(ages, using="name")`
6. Which word shows the rows with outliers? (day 19)
   → `table.show_rows_with_outliers("marks", for_each="class")`
7. How do you make 100 random numbers with average 50 and standard deviation 10, the same every time? (day 17)
   → `sdata.random_numbers(how_many=100, average=50, standard_deviation=10, seed=1)`
8. Which option makes a bar show the median instead of the average? (day 8)
   → `show="median"`
9. How do you write the text Asha in Python? (day 1)
   → `"Asha"`, in quotes. Without quotes, Python thinks Asha is a name.

Task 1. School table: the average `"marks"` for each value of `"breakfast"`, a test of the difference, and boxes of the marks for each group.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.average_of("marks", for_each="breakfast")
table.compare_averages("marks", for_each="breakfast")
chart = table.chart()
chart.boxes(x="breakfast", y="marks")
chart.save("breakfast.png")
```

A summary, a test and a chart of the same question.

Back to day 21

Day 22: Models: explain one column from several

1. Which word compares one average with a number you choose? (day 20)
   → `table.compare_average_to("marks", 60)`
2. Which word checks whether values are bell-shaped (normal)? (day 19)
   → `table.compare_shape_to("marks", "normal")`
3. Which word adds a new column made from other columns? (day 15)
   → `table.add_column("total", "Maths + Science")`, or `table.add_column("per_hour", by_dividing=["marks", "hours"])`
4. Which option adds error bars that say how sure we are of the average? (day 8)
   → `error="standard error"`, or `error="confidence interval"` for the 95 % interval.

Task 1. School table: explain `"days_absent"` using `"sleep"` and `"breakfast"` with `outcome="counts"`. Does breakfast matter?

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.explain("days_absent", using=["sleep", "breakfast"], outcome="counts")
```

Counts of days call for Poisson regression; read the line for breakfast and its p-value.

Back to day 22

Day 23: Patterns across many columns

1. Which word explains marks from hours, sleep and breakfast together? (day 22)
   → `table.explain("marks", using=["hours", "sleep", "breakfast"])`
2. What should a results sentence give? (day 20)
   → The summary of each group with n, the effect size (with its interval), the name of the test and the exact p-value.
3. Which word gives several summaries at once? (day 16)
   → `table.summary_of("sales", for_each="shop", show=["average", "median", "count"])`
4. Which word sets the y-axis from 0 to 100? (day 9)
   → `chart.range_y(0, 100)`

Task 1. Meadow table: add 3 clusters and count the rows of each cluster across `"field"`.

```
import simplydata as sdata
table = sdata.table(sdata.example("meadow"))
table.add_clusters(how_many=3)
table.count_rows(for_each="cluster", across="field")
```

If each cluster is mostly one field, the fields have different plants.

Back to day 23

Day 24: Survival, and planning a study

1. Which word adds the principal components PC1 and PC2? (day 23)
   → `table.add_principal_components(how_many=2)`
2. Which word lists every change you made to a table, for your methods? (day 20)
   → `table.steps()`
3. How do you get the number in row 1, column 2 of a grid? (day 17)
   → `grid.number_at(row=1, column=2)`
4. Which word puts two charts next to each other on one page? (day 10)
   → `sdata.side_by_side(a, b)`

Task 1. How many values does each group need to find a large difference (`effect_size=0.8`) with power 0.8? And with a small one (0.2)?

```
import simplydata as sdata
sdata.how_many_needed(effect_size=0.8, power=0.8)
sdata.how_many_needed(effect_size=0.2, power=0.8)
```

> About 26 for a large difference but about 394 for a small one: small effects need big studies.

Back to day 24

Day 25: Fitting, finding and solving

1. Which word draws survival curves? (day 24)
   → `chart.survival_curves(time="days", event="died", colour="treatment")`
2. How do you explain a yes-or-no column such as `"passed"`? (day 22)
   → `outcome="yes or no"` (logistic regression)
3. A p-value answers `"could it be chance?"`. What answers `"how big is it?"` (day 18)
   → The effect size: the difference in averages, Cohen's d or Hedges' g, omega-squared.
4. If `keep_rows` keeps rows, what would you guess takes columns away? (day 11)
   → `remove_columns`: verb + thing. `keep_` and `remove_` work on rows, columns and numbers.

Task 1. Growth table: fit the formula `"forest = a + b * (year - 2000)"` for each country. Which country loses forest fastest?

```
import simplydata as sdata
table = sdata.table(sdata.example("growth"))
table.fit_formula("forest = a + b * (year - 2000)", for_each="country")
```

> b is the change in forest each year: the country with the most negative b loses forest fastest.

Task 2. A box has a square bottom of side x and a height of 10 - x, so its volume is x * x * (10 - x). For which x between 0 and 10 is the volume biggest?

```
import simplydata as sdata
sdata.find_biggest("x * x * (10 - x)", between=[0, 10])
```

> x = 6.67 (that is, 20/3), with a volume of about 148.

Back to day 25

Day 26: Distributions and values between points

1. Which word fits a straight line on a table? (day 25)
   → `table.fit_line(x="hours", y="marks")`
2. Which word puts rows that are alike into groups? (day 23)
   → `table.add_clusters(how_many=2)`
3. You did many tests at once. Which word corrects the p-values? (day 19)
   → `add_adjusted_p_value_column(method="holm")`, on the table of results
4. How do you look at the first 5 rows of a table? (day 12)
   → `table.show_first_rows(5)`

Task 1. Heights are normal with average 150 and standard deviation 8. What share is taller than 165? Which height do 95 % of them stay below?

```
import simplydata as sdata
heights = sdata.distribution("normal", average=150, standard_deviation=8)
heights.chance(above=165)
heights.percentile(95)
```

> About 3 % are taller than 165, and 95 % are below about 163.

Task 2. Weather table: estimate the `"rain"` of each city at `month_number` 2.5 and 7.5, joined by straight lines.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
```

```
rain = table.values_at("rain", at={"month_number": [2.5, 7.5]}, joined_by="straight lines", for_each="city")
rain.show()
```

> Each estimate lies halfway between the two months around it.

Back to day 26

Day 27: Small project: look, clean and sum up

1. How do you make a normal distribution with average 50 and standard deviation 10? (day 26)
   → `sdata.distribution("normal", average=50, standard_deviation=10)`
2. Which word cuts a column of numbers into named groups? (day 15)
   → `table.add_group_column("grade", from_column="marks", cut_at=[40, 60, 80], names=["D", "C", "B", "A"])`
3. How do you give the panels of a figure the letters A and B? (day 20)
   → `letters=True` in `sdata.grid(...)` or `sdata.side_by_side(...)`
4. Which word tests whether survival differs between groups? (day 24)
   → `table.compare_survival(time="days", event="died", for_each="treatment")`

Task 1. Write your question. Open your file, describe its columns, deal with the empty cells and make two summaries for each group.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.describe_columns()
table.count_empty_cells()
table.remove_rows_with_empty_cells("sleep")
table.average_of("marks", for_each="breakfast")
table.summary_of("sleep", for_each="breakfast", show=["average", "count"])
```

> Question: do pupils who eat breakfast get better marks? Look, clean, sum up.

Back to day 27

Day 28: Small project: test, figure and results

1. Which word adds the values a model predicts as a new column? (day 22)
   → `table.add_predicted_column("marks", using=["hours", "breakfast"])`
2. Which option gives the rank-based test, for small groups that are not bell-shaped? (day 18)
   → `based_on="ranks"`
3. A word for one axis ends in …? (day 11)
   → _x or _y: `label_y`, `range_x`, `log_scale_y`.
4. How do you fit a curve with x and x squared? (day 25)
   → `table.fit_curve(x="hours", y="marks", degree=2)`

Task 1. Test your question, draw one figure and save it, and write two sentences of results.

```
import simplydata as sdata
table = sdata.table(sdata.example("school"))
table.compare_averages("marks", for_each="breakfast")
chart = table.chart()
chart.boxes(x="breakfast", y="marks")
chart.all_dots(x="breakfast", y="marks")
chart.label_y("Mark (out of 100)")
chart.save("breakfast.png")
```

> Two sentences: the averages of both groups, then the test's name, the difference and the p-value.

Back to day 28

Day 29: Final test from memory

1. Which word tests whether groups differ in their make-up across many columns (PERMANOVA)? (day 23)
   → `table.compare_profiles(for_each="grazed")`
2. Which word gives the chance of a value above 80? (day 26)
   → `dist.chance(above=80)`
3. Which word says how many values you need in each group? (day 24)
   → `sdata.how_many_needed(effect_size=0.5, power=0.8)`
4. Which word hands a table on to pandas? (day 20)
   → `table.as_pandas()`
5. Which word fits a formula of your own, such as growth? (day 25)
   → `table.fit_formula("population = a * exp(b * (year - 2000))")`
6. Which file endings give a picture that stays sharp at any size? (day 11)
   → `.pdf` and `.svg`
7. What does every simplydata program start with? (day 2)
   → import simplydata as sdata
8. Where in an error message do you look first? (day 3)
   → At the last line: it names the kind of error and says what went wrong.
9. How do you name the y-axis? (day 4)
   → `chart.label_y("Marks out of 100")`
10. Which word lets you choose the colours of the groups? (day 5)
    → `chart.colours("orange", "teal", "purple")`
11. Which word keeps only the rows where subject is Maths? (day 13)
    → `table.keep_rows("subject", equal_to="Maths")`
12. Which word draws one dot for each row, x against y? (day 6)
    → `chart.dots(x="hours", y="marks")`
13. How do you give the number 12 the name age? (day 1)
    → age = 12. The name goes on the left of =, the value on the right.
14. Which word draws a straight line through the dots? (day 8)
    → `chart.fit_line()`, after `chart.dots(...)`

Task 1. From memory, in an empty file: open the `"marks"` table, remove the rows with empty cells, give the average marks for each class, test the difference, and save a two-panel figure (boxes, and dots of hours against marks with a fitted line) as a PDF.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.remove_rows_with_empty_cells()
table.average_of("marks", for_each="class")
table.compare_averages("marks", for_each="class")
a = table.chart()
a.boxes(x="class", y="marks")
b = table.chart()
b.dots(x="hours", y="marks")
b.fit_line()
figure = sdata.grid(a, b, per_row=2, letters=True)
figure.save("final.pdf")
```

> If you wrote this from memory, you know the core of simplydata. Words you missed: run their `sdata.help` again tomorrow.

Back to day 29

Day 30: Look back, and what comes next

1. Which word fits a distribution to a column? (day 26)
   → `table.fit_distribution("days_absent", "negative binomial")`
2. Which word finds where a formula is smallest? (day 25)
   → `sdata.find_smallest("x**2 - 4*x + 1", between=[0, 5])`
3. Which word adds the principal components PC1 and PC2? (day 23)
   → `table.add_principal_components(how_many=2)`

4. Which word gives the average of a column for each group? (day 16)
   → `table.average_of("marks", for_each="class")`

Task 1. Do the task of your hardest day again from memory. Then hand the marks table to pandas and print its first rows with pandas' own word `head()`.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
df = table.as_pandas()
print(df.head())
```

> `head()` is pandas' word for `show_first_rows()`. simplydata knows many such names: `sdata.help("head")` points to the simplydata word.

Back to day 30